🏢 Recommender System for Personalized Financial Advice

Powered by PySpark + Federated Learning (Flower) + Gradio/Streamlit Deployment

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Dataset: Santander Product Recommendation — Kaggle
Tools: Google Colab · PySpark · Scikit-learn · Flower (flwr) · Gradio
Model: Logistic Regression (OneVsRest) with FedAvg Federated Learning

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1. 📋 Business Understanding

1.1 Background

Financial institutions like Santander Bank serve millions of customers with diverse financial needs. Traditional advisory services are costly, inconsistent, and inaccessible to many. A Recommender System can automate personalised product suggestions by learning from customer behaviour — improving satisfaction and increasing cross-sell revenue.

1.2 Business Problem

> *How can Santander Bank automatically suggest the right financial products to the right customers, at the right time — while keeping customer data private?*

1.3 Business Objectives

1. Build a system that recommends personalised financial products based on customer profiles
2. Use 1.5 years of historical behaviour data to learn product ownership patterns
3. Protect customer privacy using **Federated Learning** — no raw data is shared
4. Deploy the system as a user-friendly interface for customers or bank staff

1.4 Research Questions

#	Research Question
Q1	What financial products are most commonly owned by Santander customers?
Q2	How does customer age affect financial product ownership?
Q3	Does income level influence which products a customer holds?
Q4	Which customer segments represent the best cross-selling opportunities?
Q5	Can a federated learning model accurately predict product recommendations?
Q6	How do the 3 federated clients (age groups) compare in model performance?

1.5 Success Criteria

- Recommendation model achieves **F1 Score > 0.70** across all client groups
- Federated learning successfully simulated across **3 client groups**
- A working **deployment UI** is built for end-user interaction
- Recommendations are **explainable** and meaningful to non-technical users

1.6 Why Federated Learning?

In real banking environments, customer data is **highly regulated** (GDPR, CCPA) and cannot be centralised. Federated Learning solves this by training models **locally** on each client's private data — only model weights (not raw data) are ever shared.

In this project, we simulate this by splitting customers into **3 age-based client groups**, each training independently before aggregating weights on a central server.

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2. 📂 Data Understanding

2.1 Dataset Overview

- **Source:** Santander Product Recommendation — Kaggle
- **Full Size:** ~13 million rows | We use **50,000** for Colab performance

- **Time Period:** January 2015 – June 2016 (1.5 years of monthly snapshots)
- **Format:** CSV with one row per customer per month

2.2 Column Groups

Customer Profile (Features — inputs to the model):

Original Name	Renamed To	Description
ncodpers	customer_id	Unique customer ID
sexo	gender	Customer gender
age	age	Customer age
renta	household_income	Gross household income
segmento	customer_segment	01-VIP, 02-Individuals, 03-University
antiguedad	seniority_months	Months as a bank customer
ind_nuevo	is_new_customer	1 if joined in last 6 months
ind_actividad_cliente	is_active_customer	1=Active, 0=Inactive

Financial Products (Targets — what we predict, columns 25–48):

Original Name	Renamed To	Product
ind_cco_fin_ult1	has_current_account	Current Account
ind_tjcr_fin_ult1	has_credit_card	Credit Card
ind_hip_fin_ult1	has_mortgage	Mortgage
ind_plan_fin_ult1	has_pension_plan	Pension Plan
ind_ahor_fin_ult1	has_savings_account	Savings Account
ind_ecue_fin_ult1	has_eaccount	Online Account
...	...	24 product columns total

2.3 Known Data Challenges

- All columns load as **strings** due to `'NA'` values instead of proper NULLs
- Income column (`renta`) has a suspicious spike at ~€120,000 — likely imputed defaults
- Some product columns have **near-zero ownership** (class imbalance)
- Test data has **no product columns** — used only for deployment demo

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3. ⚙️ Environment Setup

3.1 Mount Google Drive

Connect Google Drive to access the dataset stored in Drive storage.

```
# Mount Google Drive
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

3.2 Install Required Libraries

Install PySpark for big data processing and Flower for federated learning.

```
# Install PySpark (big data) and Flower (federated learning)
!pip install flwr pyspark -q
print("✅ Libraries installed!")
```

```
✅ Libraries installed!
```

3.3 Import Libraries

Import all libraries needed throughout the project.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from pyspark.sql import functions as F
```

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4. 📥 Data Loading

4.1 Initialize PySpark and Load Dataset

PySpark enables distributed processing of large datasets.
We limit to 50,000 rows for Colab compatibility — sufficient for a school project.

```
from pyspark.sql import SparkSession

# Start Spark session
spark = SparkSession.builder.appName("FinancialRecommender").getOrCreate()

# Load data
df = spark.read.csv('/content/drive/MyDrive/train_ver2.csv',
                            header=True, inferSchema=True).limit(50000)
```

4.2 Statistical Summary

Generate descriptive statistics to understand value ranges and spot anomalies.

```
df.describe().show()
```

```
+-------+-----------------+------------+---------------+-----+------------------+--------------------+-----------------+--------
|summary|         ncodpers|ind_empleado|pais_residencia| sexo|               age|           ind_nuevo|       antiguedad|
+-------+-----------------+------------+---------------+-----+------------------+--------------------+-----------------+--------
|  count|            50000|       49673|          49673|49673|             50000|               50000|            50000|
|   mean|    1056305.54278|        NULL|           NULL| NULL|28.731926801280373|2.214482716968977E-4|34.33646045135184|1.138103
| stddev|29736.84503542207|        NULL|           NULL| NULL|10.340133032467396|0.014879640102633393|2.611392309743793| 3.67631
|    min|        1013024.0|           N|             AT|    H|                 4|                   0|                0|
|    max|        1375586.0|           N|             NL|    V|                NA|                  NA|              157|
+-------+-----------------+------------+---------------+-----+------------------+--------------------+-----------------+--------
```

4.3 Dataset Dimensions

Confirm how many rows and columns were loaded successfully.

```
print(f"Number of rows: {df.count()}")
print(f"Number of columns: {len(df.columns)}")
```

```
Number of rows: 50000
Number of columns: 48
```

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5. 🔍 Data Quality Assessment

5.1 Check for Duplicate Rows

Duplicates can bias model training by over-representing certain patterns.

```
# Check for duplicates
duplicate_rows_count = df.groupBy(df.columns).count().filter("count > 1").count()
print(f"Number of duplicate rows: {duplicate_rows_count}")
```

```
Number of duplicate rows: 0
```

5.2 Check for Missing Values

Count NULL and empty string values per column.
This determines which columns need imputation and which should be dropped.

```
# 2. Count nulls + empty strings per column based on column type
missing_counts_expressions = []
for column_name, data_type in df.dtypes:
    if data_type == 'string':
        # For string columns, check for nulls and empty strings
        missing_counts_expressions.append(
            F.count(F.when(F.col(column_name).isNull() | (F.col(column_name) == ""), column_name)).alias(column_name)
        )
    else:
        # For non-string columns, only check for nulls
        missing_counts_expressions.append(
            F.count(F.when(F.col(column_name).isNull(), column_name)).alias(column_name)
        )

missing_counts = df.select(missing_counts_expressions)
missing_counts.show()
```

```
+----------+--------+------------+---------------+----+---+----------+---------+----------+------+--------------+-----------+---
|fecha_dato|ncodpers|ind_empleado|pais_residencia|sexo|age|fecha_alta|ind_nuevo|antiguedad|indrel|ult_fec_cli_1t|indrel_1mes|tip
+----------+--------+------------+---------------+----+---+----------+---------+----------+------+--------------+-----------+---
|         0|       0|         327|            327| 327|  0|       327|        0|         0|     0|         49930|        327|
+----------+--------+------------+---------------+----+---+----------+---------+----------+------+--------------+-----------+---
```

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6. 🧹 Data Cleaning

6.1 Drop High-Missing Columns

Two columns are dropped due to excessive missing values:

- `ult_fec_cli_1t` — 49,930 missing (~100%)
- `conyuemp` — 50,000 missing (completely empty)

```
# Drop columns with too many missing values
df = df.drop('ult_fec_cli_1t', 'conyuemp')
```

6.2 Drop Rows with Missing Critical Profile Fields

Rows missing key profile fields (gender, segment, employment) are removed.
These are essential for content-based filtering and cannot be reliably imputed.

```
# 3. Drop rows with missing values in important columns
df = df.dropna(subset=['ind_empleado', 'sexo', 'segmento',
                       'pais_residencia', 'indrel_1mes',
                       'tiprel_1mes', 'indresi', 'indext',
                       'canal_entrada', 'indfall', 'nomprov'])
```

6.3 Impute Missing Income Values

The `renta` (income) column has ~9,154 missing values.
These are filled with the **column mean** — a standard imputation strategy that preserves the overall income distribution without introducing bias.

```
# Fill missing income (renta) with average
avg_renta = df.select(F.mean(F.col('renta'))).first()[0]
df = df.fillna({'renta': avg_renta})
```

6.4 Verify Cleaning Results

Confirm zero missing values remain across all columns.

```
# Confirm
print("Shape after cleaning:")
print((df.count(), len(df.columns)))
print("\nRemaining missing values:")
df.select([
    F.count(F.when(F.col(c).isNull(), c)).alias(c)
    for c in df.columns
]).show(vertical=True)
```

```
Shape after cleaning:
(49656, 46)

Remaining missing values:
-RECORD 0--------------------
 fecha_dato            | 0
 ncodpers              | 0
 ind_empleado          | 0
 pais_residencia       | 0
 sexo                  | 0
 age                   | 0
 fecha_alta            | 0
 ind_nuevo             | 0
 antiguedad            | 0
 indrel                | 0
 indrel_1mes           | 0
 tiprel_1mes           | 0
 indresi               | 0
 indext                | 0
 canal_entrada         | 0
 indfall               | 0
 tipodom               | 0
 cod_prov              | 0
 nomprov               | 0
 ind_actividad_cliente | 0
 renta                 | 0
 segmento              | 0
 ind_ahor_fin_ult1     | 0
 ind_aval_fin_ult1     | 0
 ind_cco_fin_ult1      | 0
 ind_cder_fin_ult1     | 0
 ind_cno_fin_ult1      | 0
 ind_ctju_fin_ult1     | 0
 ind_ctma_fin_ult1     | 0
 ind_ctop_fin_ult1     | 0
 ind_ctpp_fin_ult1     | 0
 ind_deco_fin_ult1     | 0
 ind_deme_fin_ult1     | 0
 ind_dela_fin_ult1     | 0
 ind_ecue_fin_ult1     | 0
 ind_fond_fin_ult1     | 0
 ind_hip_fin_ult1      | 0
 ind_plan_fin_ult1     | 0
 ind_pres_fin_ult1     | 0
 ind_reca_fin_ult1     | 0
 ind_tjcr_fin_ult1     | 0
 ind_valo_fin_ult1     | 0
 ind_viv_fin_ult1      | 0
 ind_nomina_ult1       | 0
 ind_nom_pens_ult1     | 0
 ind_recibo_ult1       | 0
```

6.5 Rename Columns to Readable English Names

Original Spanish column names and cryptic codes are renamed to descriptive English labels for clarity in analysis, visualisations, and the deployment UI.

```
# Rename columns to readable English names
rename_map = {
```

```
    'fecha_dato': 'record_date',
    'ncodpers': 'customer_id',
    'ind_empleado': 'employee_status',
    'pais_residencia': 'country_of_residence',
    'sexo': 'gender',
    'age': 'age',
    'fecha_alta': 'customer_since',
    'ind_nuevo': 'is_new_customer',
    'antiguedad': 'seniority_months',
    'indrel': 'customer_priority',
    'indrel_1mes': 'customer_type_month_start',
    'tiprel_1mes': 'relationship_type_month_start',
    'indresi': 'is_resident',
    'indext': 'is_foreigner',
    'canal_entrada': 'join_channel',
    'indfall': 'is_deceased',
    'tipodom': 'address_type',
    'cod_prov': 'province_code',
    'nomprov': 'province_name',
    'ind_actividad_cliente': 'is_active_customer',
    'renta': 'household_income',
    'segmento': 'customer_segment',
    'ind_ahor_fin_ult1': 'has_savings_account',
    'ind_aval_fin_ult1': 'has_guarantees',
    'ind_cco_fin_ult1': 'has_current_account',
    'ind_cder_fin_ult1': 'has_derivada_account',
    'ind_cno_fin_ult1': 'has_payroll_account',
    'ind_ctju_fin_ult1': 'has_junior_account',
    'ind_ctma_fin_ult1': 'has_mas_particular_account',
    'ind_ctop_fin_ult1': 'has_particular_account',
    'ind_ctpp_fin_ult1': 'has_particular_plus_account',
    'ind_deco_fin_ult1': 'has_short_term_deposit',
    'ind_deme_fin_ult1': 'has_medium_term_deposit',
    'ind_dela_fin_ult1': 'has_long_term_deposit',
    'ind_ecue_fin_ult1': 'has_eaccount',
    'ind_fond_fin_ult1': 'has_funds',
    'ind_hip_fin_ult1': 'has_mortgage',
    'ind_plan_fin_ult1': 'has_pension_plan',
    'ind_pres_fin_ult1': 'has_loan',
    'ind_reca_fin_ult1': 'has_taxes',
    'ind_tjcr_fin_ult1': 'has_credit_card',
    'ind_valo_fin_ult1': 'has_securities',
    'ind_viv_fin_ult1': 'has_home_account',
    'ind_nomina_ult1': 'has_payroll',
    'ind_nom_pens_ult1': 'has_pension',
    'ind_recibo_ult1': 'has_direct_debit'
}

# Apply renaming in PySpark
for old_name, new_name in rename_map.items():
    df = df.withColumnRenamed(old_name, new_name)

# Confirm
print("New column names:")
print(df.columns)
```

```
New column names:
['record_date', 'customer_id', 'employee_status', 'country_of_residence', 'gender', 'age', 'customer_since', 'is_new_customer',
```

6.6 Preview Cleaned Dataset

Verify the renaming worked correctly by displaying key columns.

```
df.select('customer_id', 'age', 'household_income',
          'customer_segment', 'has_current_account',
          'has_credit_card', 'has_mortgage').show(5)
```

```
+-----------+---+------------------+------------------+-------------------+---------------+------------+
|customer_id|age|  household_income|  customer_segment|has_current_account|has_credit_card|has_mortgage|
+-----------+---+------------------+------------------+-------------------+---------------+------------+
|  1375586.0| 35|           87218.1| 02 - PARTICULARES|                  1|              0|           0|
|  1050611.0| 23|          35548.74|03 - UNIVERSITARIO|                  1|              0|           0|
|  1050612.0| 23|122179.11000000002|03 - UNIVERSITARIO|                  1|              0|           0|
|  1050613.0| 22|         119775.54|03 - UNIVERSITARIO|                  0|              0|           0|
|  1050614.0| 23|116040.98494528145|03 - UNIVERSITARIO|                  1|              0|           0|
```

```
+-----------+---+------------------+------------------+------------------+---------------+------------+
only showing top 5 rows
```

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7. 📊 Exploratory Data Analysis (EDA)

EDA answers our research questions through visualisation before modelling begins.

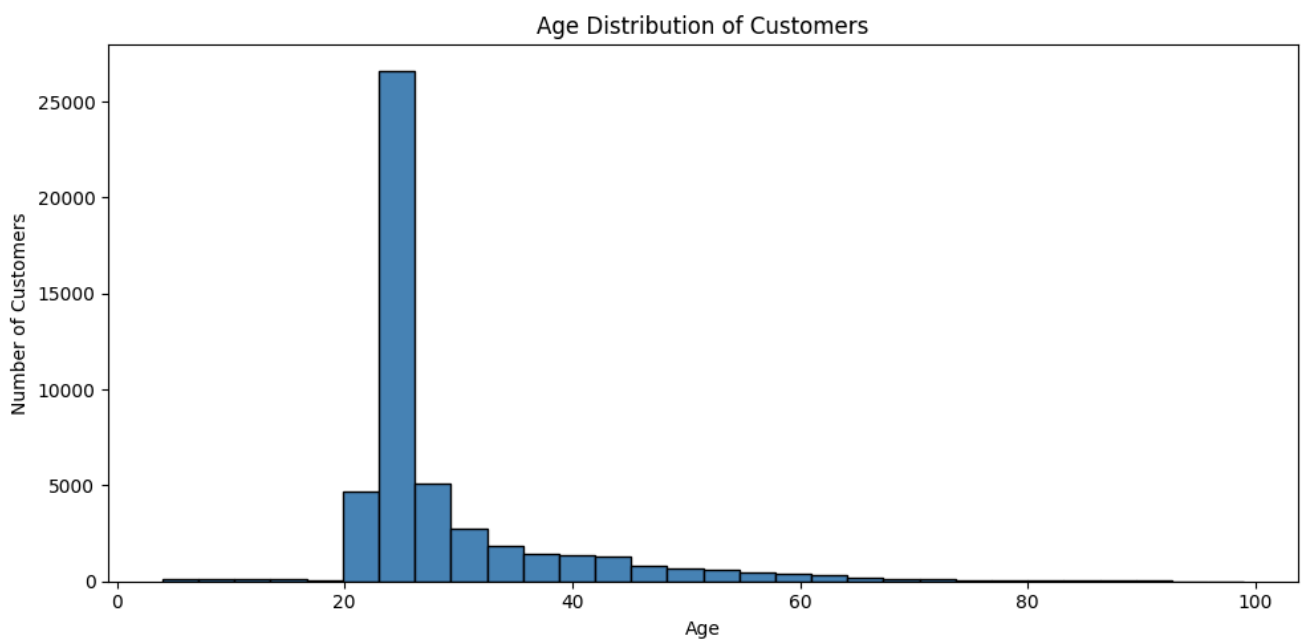
Questions addressed:

- Q1: Most common products?
- Q2: How does age affect ownership?
- Q3: Does income predict product choice?
- Q4: Which segments have the most cross-sell potential?

7.1 Age Distribution (Q2)

Understand the age spread of the customer base.

```
# Convert age column to pandas for plotting
age_df = df.select('age').toPandas()
age_df['age'] = pd.to_numeric(age_df['age'], errors='coerce')

plt.figure(figsize=(10, 5))
plt.hist(age_df['age'].dropna(), bins=30, color='steelblue', edgecolor='black')
plt.title('Age Distribution of Customers')
plt.xlabel('Age')
plt.ylabel('Number of Customers')
plt.tight_layout()
plt.show()
```



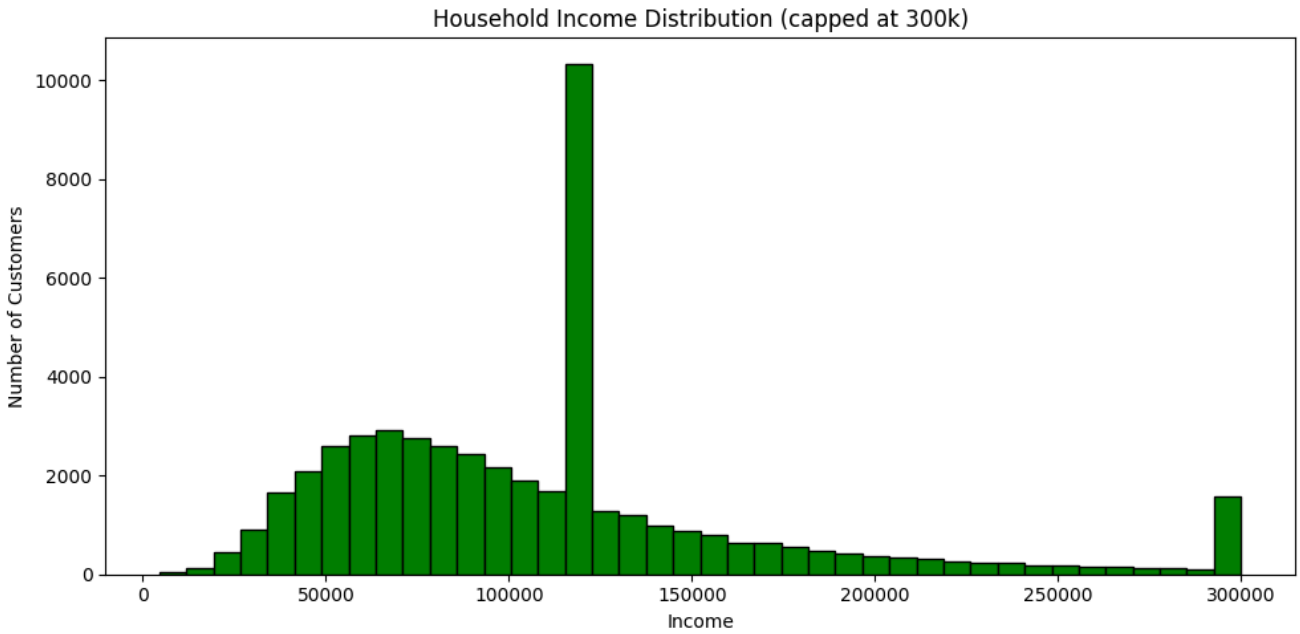
- The majority of customers are aged 20–30, with a sharp peak around 23–25
- This suggests the dataset is dominated by young/university-aged customers
- Very few customers above 60 — good insight for tailoring recommendations by age group

Key Finding (Q2): The majority of customers are aged **20–30**, peaking at 23–25. This dataset is dominated by young/university-aged customers. Very few customers above 60 — age is a strong segmentation signal for recommendations.

7.2 Household Income Distribution (Q3)

Identify income patterns and potential imputation artefacts.

```
income_df = df.select('household_income').toPandas()
income_df['household_income'] = pd.to_numeric(income_df['household_income'], errors='coerce')

plt.figure(figsize=(10, 5))
plt.hist(income_df['household_income'].dropna().clip(upper=300000),
         bins=40, color='green', edgecolor='black')
plt.title('Household Income Distribution (capped at 300k)')
plt.xlabel('Income')
plt.ylabel('Number of Customers')
plt.tight_layout()
plt.show()
```

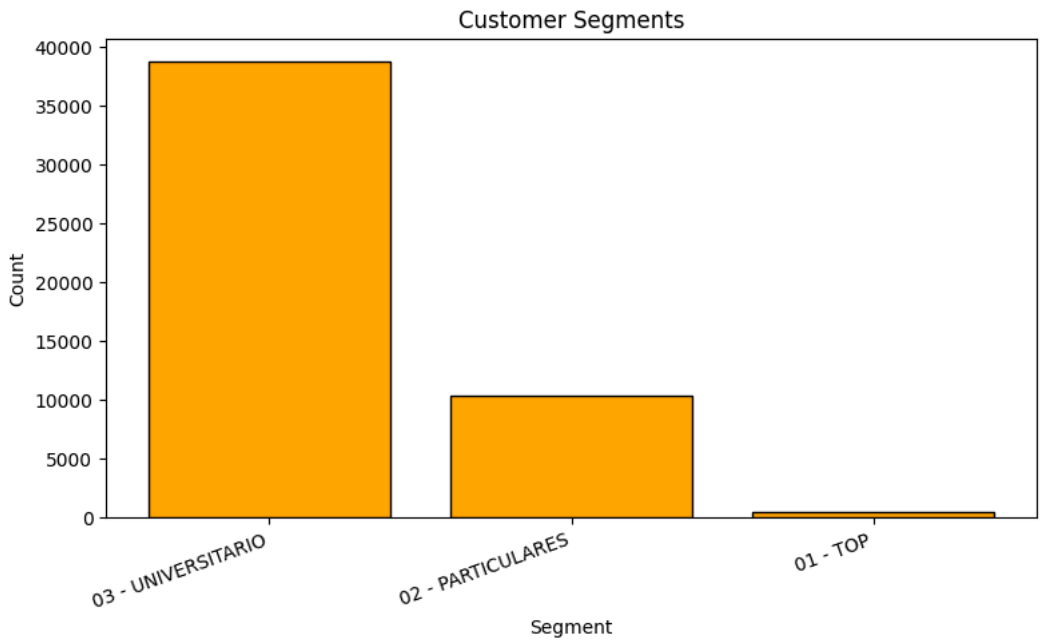


Key Finding (Q3): Most incomes fall between **€40,000–€130,000**.
The spike at €120,000 is unusual — likely a default imputed value assigned to many customers.
Income alone is a weak predictor (confirmed in the correlation analysis).

7.3 Customer Segments (Q4)

Understand the distribution of customers across the 3 bank segments.

```
segment_df = df.groupBy('customer_segment').count().orderBy('count', ascending=False).toPandas()

plt.figure(figsize=(8, 5))
plt.bar(segment_df['customer_segment'], segment_df['count'], color='orange', edgecolor='black')
plt.title('Customer Segments')
plt.xlabel('Segment')
plt.ylabel('Count')
plt.xticks(rotation=20, ha='right')
plt.tight_layout()
plt.show()
```



Key Finding (Q4):

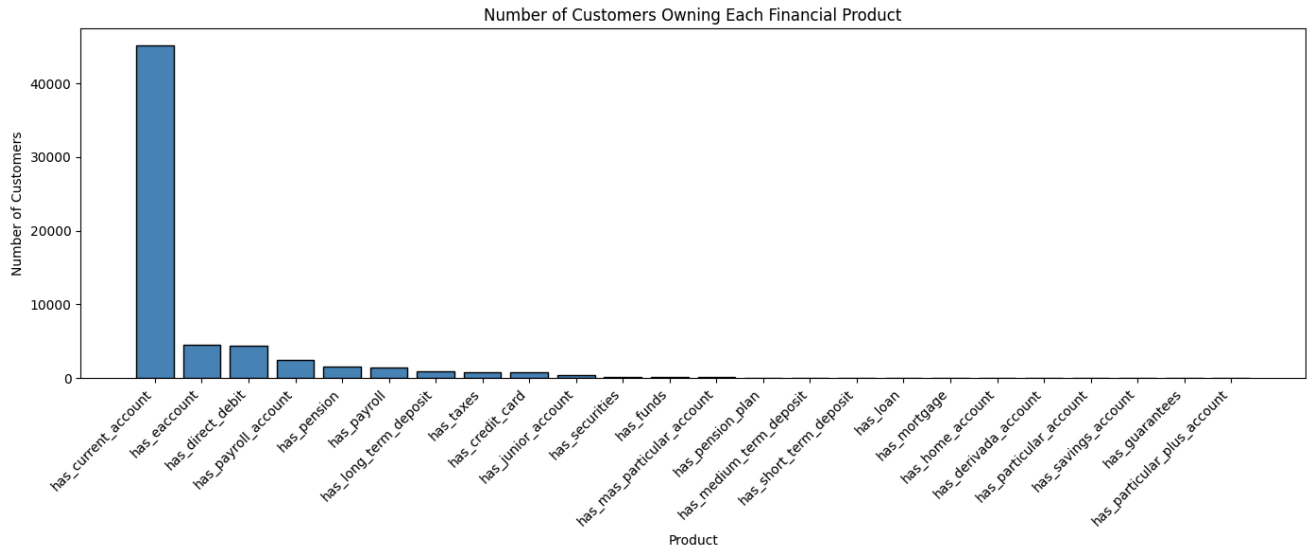
- **UNIVERSITARIO (03)** — ~78% of all customers (dominant group)
- **PARTICULARES (02)** — ~20% of customers
- **TOP/VIP (01)** — ~1% (very small but highest value)

University students represent the largest cross-sell opportunity.

7.4 Product Ownership Overview (Q1)

Count how many customers own each of the 24 financial products.

```
from pyspark.sql.functions import sum as spark_sum , col
from pyspark.sql.functions import col # Import col function

product_cols = [c for c in df.columns if c.startswith('has_')]

# Sum each product column, casting to int to handle potential string 'NA' values
product_counts = df.select([spark_sum(col(c).try_cast('int')).alias(c) for c in product_cols])
product_df = product_counts.toPandas().T.reset_index()
product_df.columns = ['product', 'count']
product_df = product_df.sort_values('count', ascending=False)

plt.figure(figsize=(14, 6))
plt.bar(product_df['product'], product_df['count'], color='steelblue', edgecolor='black')
plt.title('Number of Customers Owning Each Financial Product')
plt.xlabel('Product')
plt.ylabel('Number of Customers')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



Key Finding (Q1):

- **Current Account** — nearly universal (~45,000 customers)
- **E-account** and **Direct Debit** — next most common (~4,500 each)
- All other products — very low ownership — this is the **recommendation opportunity**

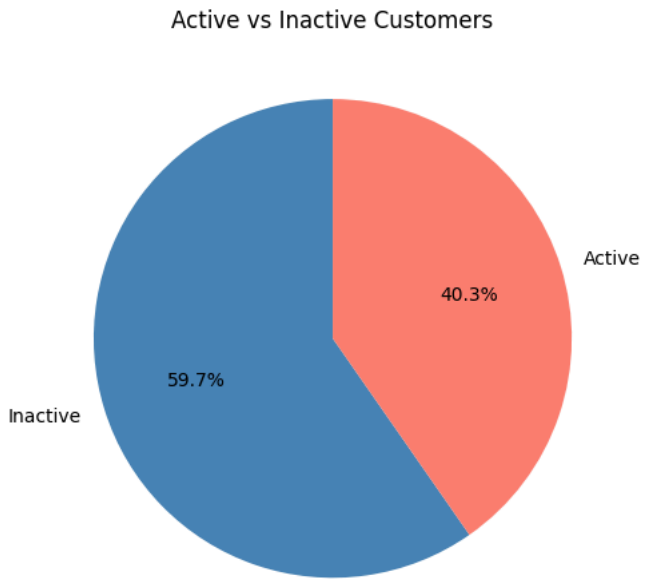
7.5 Active vs Inactive Customers

Measure engagement levels to prioritise recommendation targets.

```
# Convert to pandas and inspect what values actually exist
active_df = df.select('is_active_customer').toPandas()

# Check what unique values are in the column
print("Unique values:", active_df['is_active_customer'].unique())
print("Data type:", active_df['is_active_customer'].dtype)
```

```
Unique values: [' 1' ' 0']
Data type: object
```

```
# Force convert to integer then map
active_df['is_active_customer'] = pd.to_numeric(
    active_df['is_active_customer'], errors='coerce'
).dropna().astype(int)

counts = active_df['is_active_customer'].value_counts().reset_index()
counts.columns = ['status', 'count']
counts['status'] = counts['status'].map({1: 'Active', 0: 'Inactive'})

plt.figure(figsize=(6, 5))
plt.pie(counts['count'], labels=counts['status'],
        autopct='%1.1f%%', colors=['steelblue', 'salmon'], startangle=90)
plt.title('Active vs Inactive Customers')
plt.tight_layout()
plt.show()
```

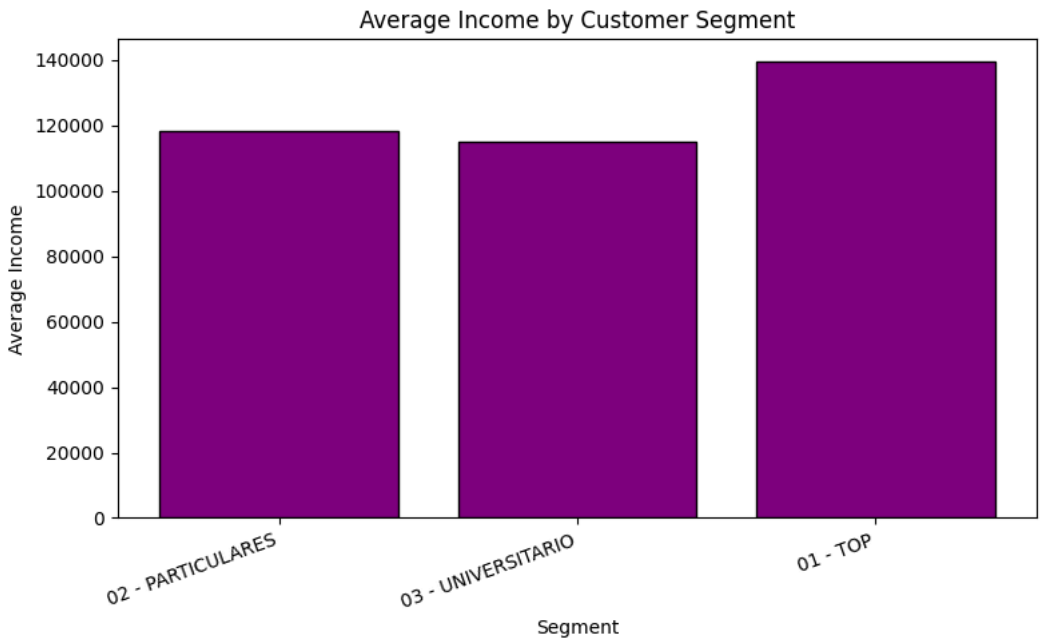


Key Finding: 59.7% of customers are inactive.
The recommender system can serve as a **re-engagement tool** — suggesting relevant products to bring inactive customers back.

7.6 Average Income by Customer Segment

Validate that segment definitions align with income levels.

```
from pyspark.sql.functions import mean as spark_mean

income_segment = df.groupBy('customer_segment') \
                   .agg(spark_mean('household_income').alias('avg_income')) \
                   .toPandas()

plt.figure(figsize=(8, 5))
plt.bar(income_segment['customer_segment'], income_segment['avg_income'],
        color='purple', edgecolor='black')
plt.title('Average Income by Customer Segment')
plt.xlabel('Segment')
plt.ylabel('Average Income')
plt.xticks(rotation=20, ha='right')
plt.tight_layout()
plt.show()
```

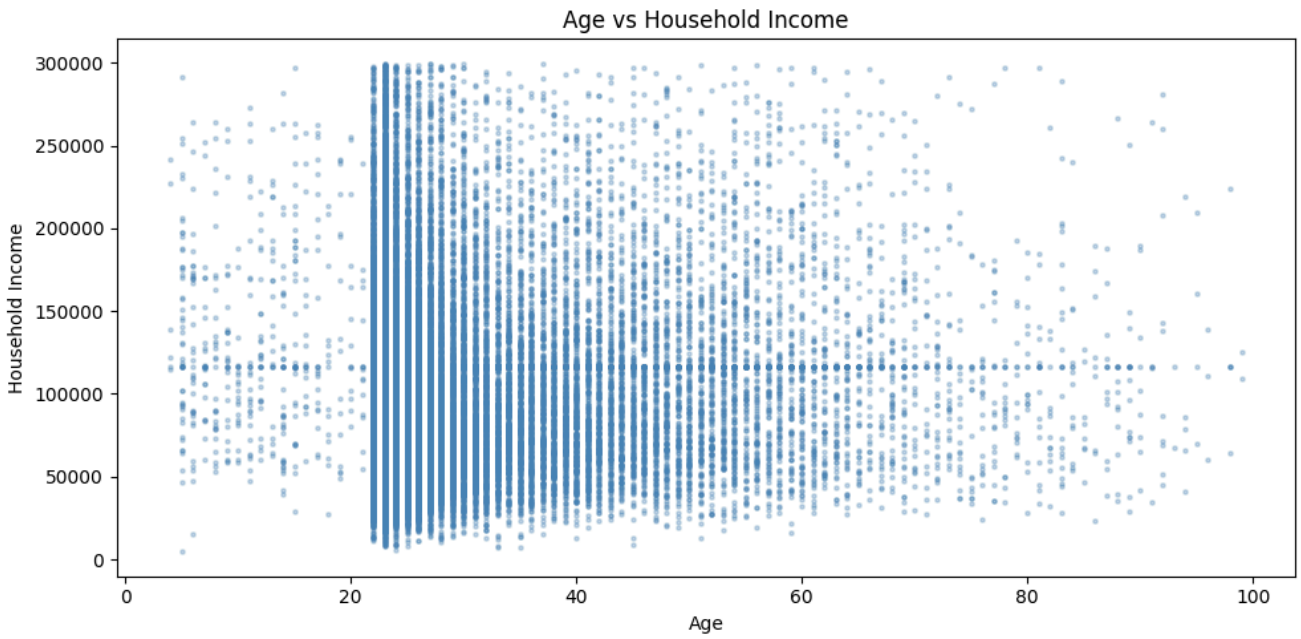


Key Finding: VIP customers earn the most (~~€140,000 avg).~~
~~University and Individual segments are similar (~~€115,000–€118,000).
Segment is a **better predictor** than income alone for recommendation targeting.

7.7 Bivariate Analysis — Age vs Household Income

Examine whether age predicts income (a key assumption for feature selection).

```
sample = df.select('age', 'household_income').toPandas().dropna()
sample['age'] = pd.to_numeric(sample['age'], errors='coerce')
sample['household_income'] = pd.to_numeric(sample['household_income'], errors='coerce')
sample = sample[(sample['age'] < 100) & (sample['household_income'] < 300000)]

plt.figure(figsize=(10, 5))
plt.scatter(sample['age'], sample['household_income'],
            alpha=0.3, color='steelblue', s=5)
plt.title('Age vs Household Income')
plt.xlabel('Age')
plt.ylabel('Household Income')
plt.tight_layout()
plt.show()
```

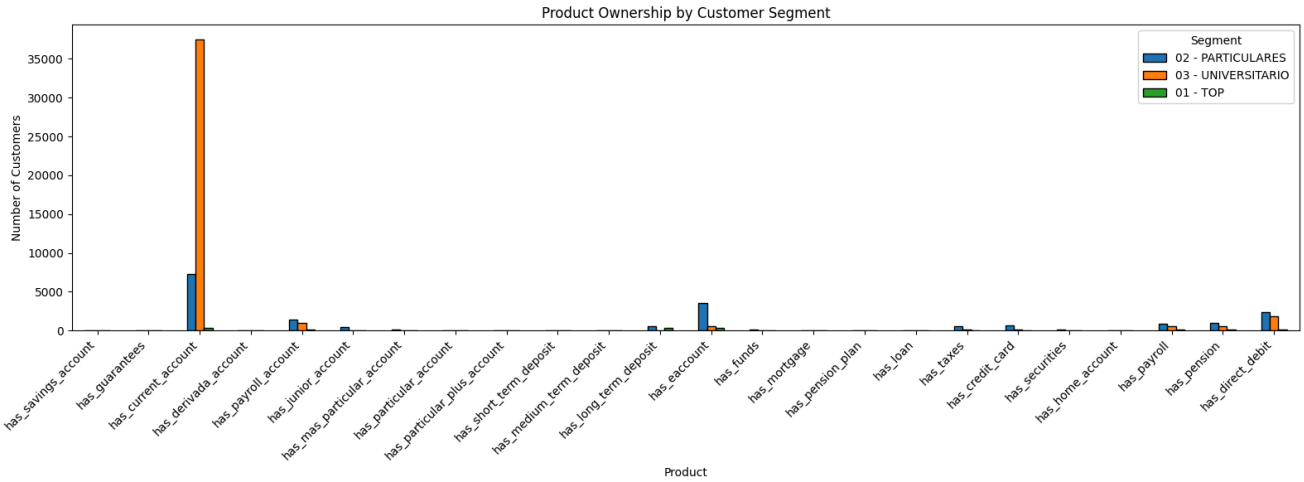


Key Finding: No strong linear relationship between age and income.
Income is widely spread across all age groups — confirming that **both features are needed together** rather than relying on either alone for recommendations.

7.8 Product Ownership by Segment (Multivariate)

Cross-tabulate segments against all 24 products to find cross-sell patterns.

```
product_cols = [c for c in df.columns if c.startswith('has_')]

segment_products = df.groupBy('customer_segment') \
    .agg(*[spark_sum(col(c).try_cast('int')).alias(c) for c in product_cols]) \
    .toPandas()

segment_products = segment_products.set_index('customer_segment').T

segment_products.plot(kind='bar', figsize=(16, 6), edgecolor='black')
plt.title('Product Ownership by Customer Segment')
plt.xlabel('Product')
plt.ylabel('Number of Customers')
plt.xticks(rotation=45, ha='right')
plt.legend(title='Segment')
plt.tight_layout()
plt.show()
```



Key Finding (Q4):

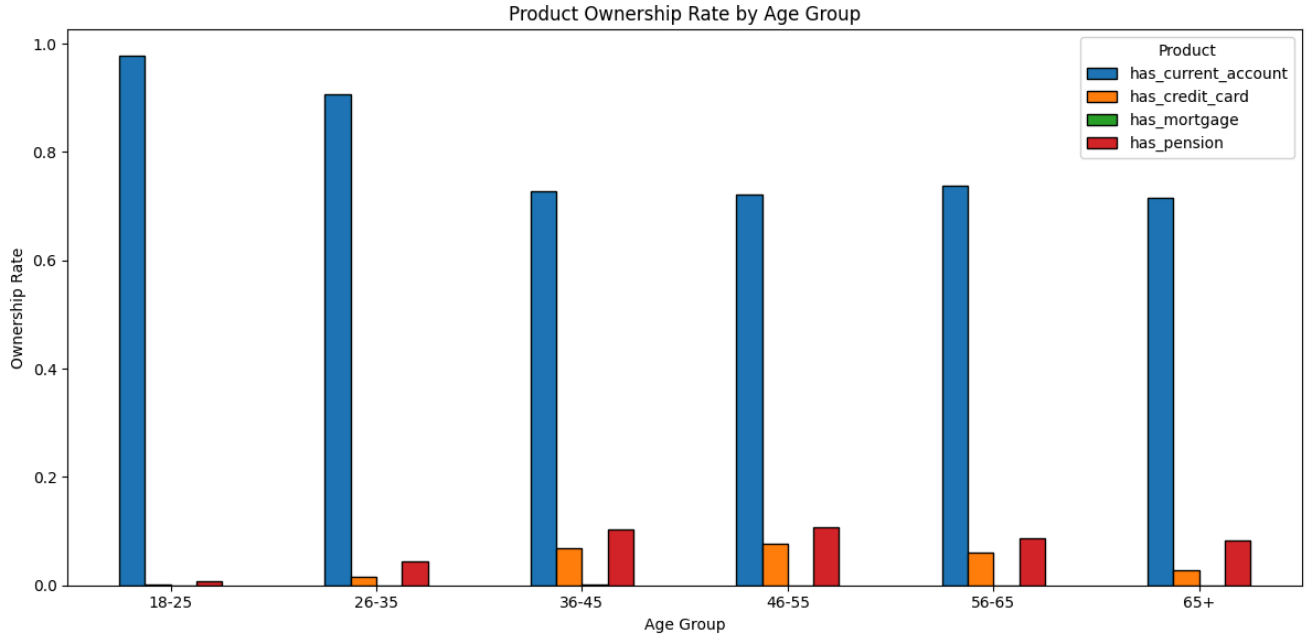
- University students mostly only have a **current account** — huge cross-selling gap
- Individuals (PARTICULARES) own a more diverse product set
- VIP group is small but should be targeted with **premium products**

7.9 Product Ownership by Age Group

Break down specific product ownership rates across age bands.

```
from pyspark.sql.functions import avg as spark_avg

# Average product ownership by age group
age_products = df.select('age', 'has_current_account', 'has_credit_card',
                         'has_mortgage', 'has_pension') \
                .toPandas()

age_products['age'] = pd.to_numeric(age_products['age'], errors='coerce')
# Ensure product columns are numeric before grouping
for col_name in ['has_current_account','has_credit_card','has_mortgage','has_pension']:
    age_products[col_name] = pd.to_numeric(age_products[col_name], errors='coerce')

age_products = age_products[age_products['age'] < 100]
age_products['age_group'] = pd.cut(age_products['age'],
                                   bins=[18, 25, 35, 45, 55, 65, 100],
                                   labels=['18-25','26-35','36-45','46-55','56-65','65+'])

age_group_products = age_products.groupby('age_group')[
    ['has_current_account','has_credit_card','has_mortgage','has_pension']
].mean()

age_group_products.plot(kind='bar', figsize=(12, 6), edgecolor='black')
plt.title('Product Ownership Rate by Age Group')
plt.xlabel('Age Group')
plt.ylabel('Ownership Rate')
plt.xticks(rotation=0)
plt.legend(title='Product')
plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_16961/2500613481.py:18: FutureWarning: The default of observed=False is deprecated and will be changed to True in
  age_group_products = age_products.groupby('age_group')[
```



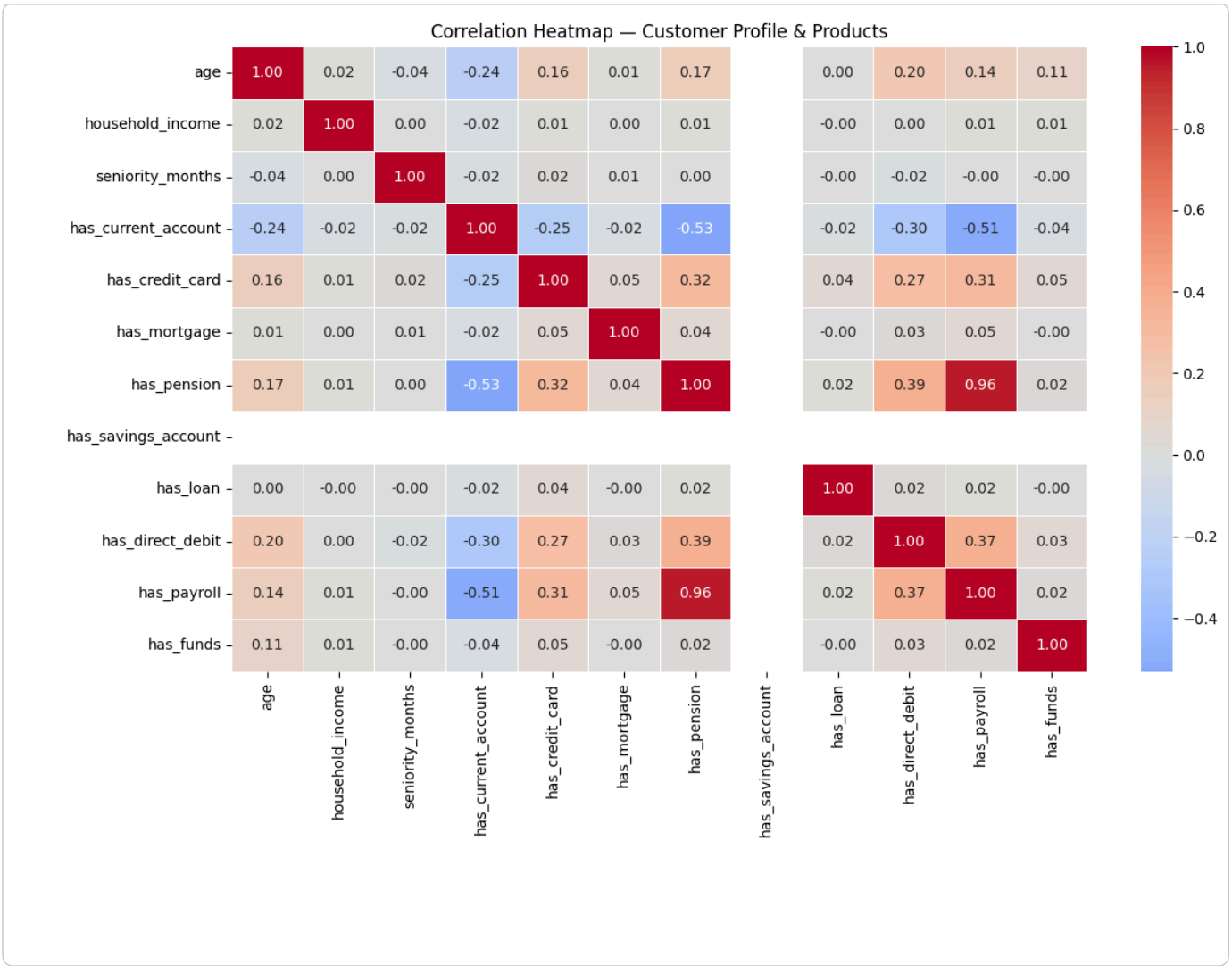
Key Finding (Q2):

- **Current account** highest among 18–25 (~98%)
- **Credit card** peaks at 36–55 — middle-aged customers are the prime credit card target
- **Pension** increases with age — confirming retirement planning behaviour
- **Mortgage** nearly invisible — very rare in this dataset

7.10 Correlation Heatmap (Multivariate)

Identify which variables are related to guide feature selection and model expectations.

```
import seaborn as sns

# Select numeric columns for correlation
corr_cols = ['age', 'household_income', 'seniority_months',
             'has_current_account', 'has_credit_card', 'has_mortgage',
             'has_pension', 'has_savings_account', 'has_loan',
             'has_direct_debit', 'has_payroll', 'has_funds']

corr_df = df.select(corr_cols).toPandas()
corr_df = corr_df.apply(pd.to_numeric, errors='coerce')

plt.figure(figsize=(12, 8))
sns.heatmap(corr_df.corr(), annot=True, fmt='.2f',
            cmap='coolwarm', center=0, linewidths=0.5)
plt.title('Correlation Heatmap — Customer Profile & Products')
plt.tight_layout()
plt.show()
```



Key Correlation Findings:

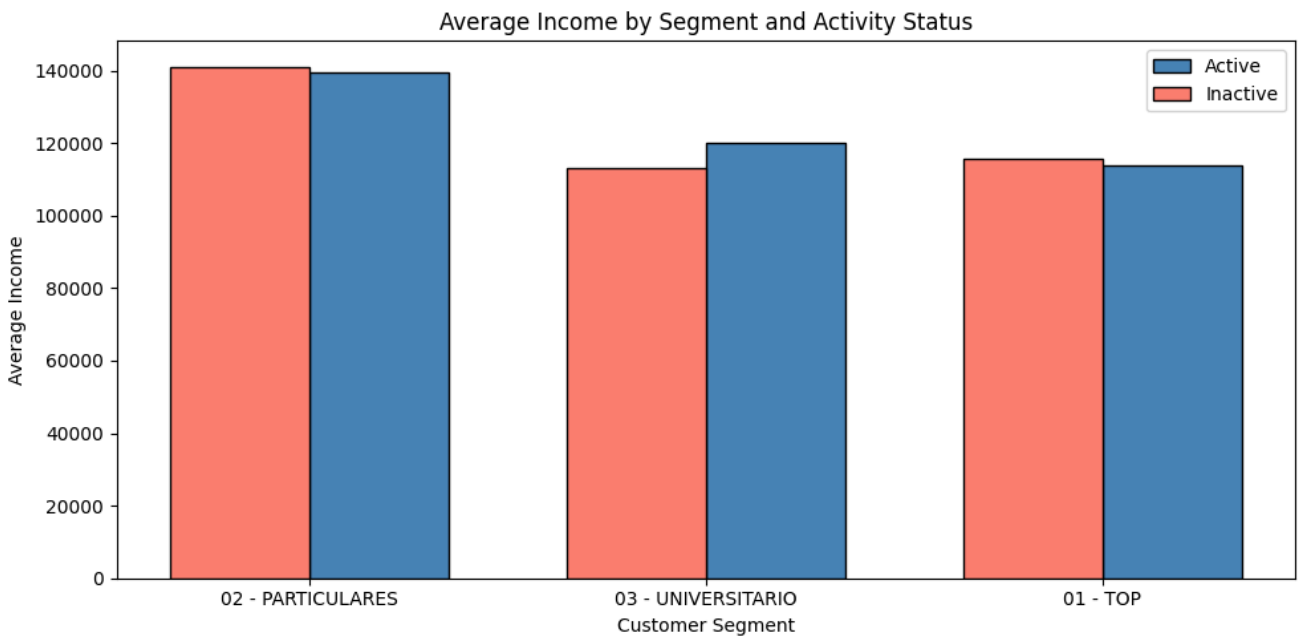
- **Pension & Payroll (0.96)** — nearly perfectly correlated → if customer has payroll, recommend pension
- **Current Account & Pension (-0.53)** — customers with only a basic account rarely have a pension
- **Income has very weak correlations** with all products → income alone is not a strong predictor
- **Age & Current Account (-0.24)** → older customers have more diverse product portfolios

7.11 Average Income by Segment and Activity Status (Multivariate)

Test whether active/inactive status interacts with income across segments.

```
income_seg = df.select('customer_segment', 'is_active_customer',
                       'household_income').toPandas()
income_seg['household_income'] = pd.to_numeric(income_seg['household_income'], errors='coerce')
income_seg['is_active_customer'] = pd.to_numeric(income_seg['is_active_customer'], errors='coerce')
income_seg['status'] = income_seg['is_active_customer'].map({1.0: 'Active', 0.0: 'Inactive'})

plt.figure(figsize=(10, 5))
for status, color in zip(['Active', 'Inactive'], ['steelblue', 'salmon']):
    subset = income_seg[income_seg['status'] == status]
    grouped = subset.groupby('customer_segment')['household_income'].mean()
    plt.bar([x + (0.35 if status == 'Active' else 0)
             for x in range(len(grouped))],
            grouped.values, width=0.35, label=status, color=color, edgecolor='black')

plt.title('Average Income by Segment and Activity Status')
plt.xlabel('Customer Segment')
plt.ylabel('Average Income')
plt.xticks([0.175, 1.175, 2.175], income_seg['customer_segment'].unique())
plt.legend()
```

```
plt.tight_layout()
plt.show()
```



Key Finding: Income differences between active and inactive customers are minimal.
Activity status is **not strongly driven by income** — other factors like product fit and engagement matter more. This validates using both features in the model.

---

8. 🔧 Data Preprocessing & Feature Engineering

Before modelling we must:

1. Fix data types (all columns loaded as strings due to 'NA' values)
2. Drop irrelevant columns (IDs, dates, geography)
3. Encode categorical text columns to numbers
4. Engineer new features to improve predictions
5. Split data into federated client groups

8.1 Drop Irrelevant Columns

Remove columns that would not help predict product ownership:

- **Identifiers** (customer_id) — would cause overfitting
- **Dates** — not in a usable format for this model
- **Geographic codes** — too many unique values (high cardinality)

```
cols_to_drop = ['record_date', 'customer_since', 'customer_id',
                'province_name', 'join_channel', 'country_of_residence',
                'address_type', 'province_code', 'is_deceased',
                'customer_priority', 'customer_type_month_start',
                'relationship_type_month_start', 'is_resident',
                'is_foreigner', 'employee_status']

df = df.drop(*cols_to_drop)
print("Remaining columns:", df.columns)
```

```
Remaining columns: ['gender', 'age', 'is_new_customer', 'seniority_months', 'is_active_customer', 'household_income', 'customer_
```

8.2 Fix Data Types

All columns loaded as strings because the dataset uses `'NA'` strings instead of NULL.
We first replace these with proper NULLs, then safely cast to numeric types.

```
from pyspark.sql.functions import col, when
from pyspark.sql.types import IntegerType, FloatType

# Replace 'NA' strings with NULL first
from pyspark.sql.functions import regexp_replace, trim

# Clean all string columns — replace 'NA', '', ' ' with null
for c in df.columns:
    df = df.withColumn(c, when(trim(col(c)) == 'NA', None)
                          .when(trim(col(c)) == '', None)
                          .otherwise(col(c)))

# Now safely cast using try_cast equivalent in PySpark
float_cols = ['age', 'household_income', 'seniority_months']

# Removed 'address_type' and 'province_code' as they were dropped previously
int_cols = ['is_new_customer', 'is_active_customer',
            'has_savings_account', 'has_guarantees',
            'has_current_account', 'has_derivada_account', 'has_payroll_account',
            'has_junior_account', 'has_mas_particular_account', 'has_particular_account',
            'has_particular_plus_account', 'has_short_term_deposit', 'has_medium_term_deposit',
            'has_long_term_deposit', 'has_eaccount', 'has_funds', 'has_mortgage',
            'has_pension_plan', 'has_loan', 'has_taxes', 'has_credit_card',
            'has_securities', 'has_home_account', 'has_payroll', 'has_pension',
            'has_direct_debit']

# Cast float columns
for c in float_cols:
    df = df.withColumn(c, col(c).try_cast(FloatType()))

# Cast int columns
for c in int_cols:
    df = df.withColumn(c, col(c).try_cast(IntegerType()))

print(" Data types fixed!")
df.printSchema()
```

```
 Data types fixed!
root
 |-- gender: string (nullable = true)
 |-- age: float (nullable = true)
 |-- is_new_customer: integer (nullable = true)
 |-- seniority_months: float (nullable = true)
 |-- is_active_customer: integer (nullable = true)
 |-- household_income: float (nullable = true)
 |-- customer_segment: string (nullable = true)
 |-- has_savings_account: integer (nullable = true)
 |-- has_guarantees: integer (nullable = true)
 |-- has_current_account: integer (nullable = true)
 |-- has_derivada_account: integer (nullable = true)
 |-- has_payroll_account: integer (nullable = true)
 |-- has_junior_account: integer (nullable = true)
 |-- has_mas_particular_account: integer (nullable = true)
 |-- has_particular_account: integer (nullable = true)
 |-- has_particular_plus_account: integer (nullable = true)
 |-- has_short_term_deposit: integer (nullable = true)
 |-- has_medium_term_deposit: integer (nullable = true)
 |-- has_long_term_deposit: integer (nullable = true)
 |-- has_eaccount: integer (nullable = true)
 |-- has_funds: integer (nullable = true)
 |-- has_mortgage: integer (nullable = true)
 |-- has_pension_plan: integer (nullable = true)
 |-- has_loan: integer (nullable = true)
 |-- has_taxes: integer (nullable = true)
 |-- has_credit_card: integer (nullable = true)
 |-- has_securities: integer (nullable = true)
 |-- has_home_account: integer (nullable = true)
 |-- has_payroll: integer (nullable = true)
 |-- has_pension: integer (nullable = true)
 |-- has_direct_debit: integer (nullable = true)
```

8.3 Encode Categorical Columns

Machine learning models require numeric inputs.

`gender` and `customer_segment` are encoded using PySpark's **StringIndexer**:

- `'Male'` → 0, `'Female'` → 1
- `'03 - UNIVERSITARIO'` → 0, `'02 - PARTICULARES'` → 1, `'01 - TOP'` → 2

```
from pyspark.ml.feature import StringIndexer
from pyspark.ml import Pipeline

categorical_cols = ['gender', 'customer_segment']

indexers = [StringIndexer(inputCol=c, outputCol=c + '_idx', handleInvalid='keep')
            for c in categorical_cols]

pipeline = Pipeline(stages=indexers)
df = pipeline.fit(df).transform(df)
df = df.drop(*categorical_cols)

print("Encoding done!")
df.show(3)
```

```
Encoding done!
+----+---------------+----------------+------------------+----------------+-------------------+--------------+------------------
| age|is_new_customer|seniority_months|is_active_customer|household_income|has_savings_account|has_guarantees|has_current_accoun
+----+---------------+----------------+------------------+----------------+-------------------+--------------+------------------
|35.0|              0|             6.0|                 1|         87218.1|                  0|             0|
|23.0|              0|            35.0|                 0|        35548.74|                  0|             0|
|23.0|              0|            35.0|                 0|       122179.11|                  0|             0|
+----+---------------+----------------+------------------+----------------+-------------------+--------------+------------------
only showing top 3 rows
```

8.4 Feature Engineering

Two new features are created to improve model performance:

1. **total_products_owned** — how many products a customer currently has
   (indicates how financially engaged the customer is)
2. **is_high_earner** — binary flag for income > €150,000
   (high earners may qualify for premium products)

```
product_cols = [c for c in df.columns if c.startswith('has_')]

# Total products owned per customer
df = df.withColumn('total_products_owned',
                   sum([col(c) for c in product_cols]))

# High earner flag
df = df.withColumn('is_high_earner',
                   (col('household_income') > 150000).cast('int'))

print("Feature engineering done!")
df.select('total_products_owned', 'is_high_earner').show(5)
```

```
Feature engineering done!
+--------------------+--------------+
|total_products_owned|is_high_earner|
+--------------------+--------------+
|                   1|             0|
|                   1|             0|
|                   1|             0|
|                   1|             0|
|                   1|             0|
+--------------------+--------------+
only showing top 5 rows
```

8.5 Convert to Pandas and Create Federated Client Splits

PySpark DataFrames are converted to Pandas for compatibility with Flower and Scikit-learn.
We split into **3 age-based federated client groups** to simulate separate bank branches:

Client	Age Range	Rationale
Client 1	age ≤ 30	Young / University customers
Client 2	30 < age ≤ 50	Middle-aged professionals
Client 3	age > 50	Older / Pre-retirement customers

```
# Convert full dataframe to Pandas
df_pandas = df.toPandas()

# Fill any remaining nulls with 0
df_pandas = df_pandas.fillna(0)

print("Total records:", len(df_pandas))
print("Columns:", df_pandas.columns.tolist())

# Split into 3 clients by age group
client_1 = df_pandas[df_pandas['age'] <= 30]
client_2 = df_pandas[(df_pandas['age'] > 30) & (df_pandas['age'] <= 50)]
client_3 = df_pandas[df_pandas['age'] > 50]

print(f"\nClient 1 (age ≤ 30) : {len(client_1)} customers")
print(f"Client 2 (age 30-50): {len(client_2)} customers")
print(f"Client 3 (age > 50) : {len(client_3)} customers")
```

```
Total records: 49656
Columns: ['age', 'is_new_customer', 'seniority_months', 'is_active_customer', 'household_income', 'has_savings_account', 'has_gu

Client 1 (age ≤ 30) : 37995 customers
Client 2 (age 30-50): 8934 customers
Client 3 (age > 50) : 2727 customers
```

8.6 Extract Features and Labels for Each Client

Define which columns are **inputs (X)** and which are **targets (y)** for each client group.

```
import numpy as np

# Features (input to model)
feature_cols = ['age', 'household_income', 'seniority_months',
                'is_active_customer', 'is_new_customer',
                'gender_idx', 'customer_segment_idx',
                'total_products_owned', 'is_high_earner']

# Targets (what we want to recommend)
product_cols = [c for c in df_pandas.columns if c.startswith('has_')]

# Extract X and y for each client
X1, y1 = client_1[feature_cols].values, client_1[product_cols].values
X2, y2 = client_2[feature_cols].values, client_2[product_cols].values
X3, y3 = client_3[feature_cols].values, client_3[product_cols].values

print("Client 1 → Features:", X1.shape, "| Products:", y1.shape)
print("Client 2 → Features:", X2.shape, "| Products:", y2.shape)
print("Client 3 → Features:", X3.shape, "| Products:", y3.shape)
print("\n All 3 clients ready for Federated Learning!")
```

```
Client 1 → Features: (37995, 9) | Products: (37995, 24)
Client 2 → Features: (8934, 9) | Products: (8934, 24)
Client 3 → Features: (2727, 9) | Products: (2727, 24)

 All 3 clients ready for Federated Learning!
```

---

9. 🌐 Federated Learning Setup

9.1 Install Flower

Flower (flwr) is the federated learning framework we use to simulate privacy-preserving distributed training across the 3 client groups.

```
!pip install flwr -q
print("Flower installed!")
```

```
Flower installed!
```

9.2 Import Federated Learning Libraries

```
import flwr as fl
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.multiclass import OneVsRestClassifier
from sklearn.metrics import precision_score, recall_score, f1_score
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
import warnings
warnings.filterwarnings('ignore')
```

9.3 Prepare Train/Test Splits and Scale Features

Each client's data is split into 80% training and 20% test sets.
`StandardScaler` normalises features to mean=0, std=1 — essential for Logistic Regression to converge correctly.

```
def prepare_client(X, y):
    """Split and scale data for a client"""
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42
    )
    # Scale features
    scaler = StandardScaler()
    X_train = scaler.fit_transform(X_train)
    X_test = scaler.transform(X_test)

    return X_train, X_test, y_train, y_test

# Prepare all 3 clients
X1_train, X1_test, y1_train, y1_test = prepare_client(X1, y1)
X2_train, X2_test, y2_train, y2_test = prepare_client(X2, y2)
X3_train, X3_test, y3_train, y3_test = prepare_client(X3, y3)

print("Client 1 → Train:", X1_train.shape, "| Test:", X1_test.shape)
print("Client 2 → Train:", X2_train.shape, "| Test:", X2_test.shape)
print("Client 3 → Train:", X3_train.shape, "| Test:", X3_test.shape)
print(" All clients prepared!")
```

```
Client 1 → Train: (30396, 9) | Test: (7599, 9)
Client 2 → Train: (7147, 9) | Test: (1787, 9)
Client 3 → Train: (2181, 9) | Test: (546, 9)
 All clients prepared!
```

---

10. Model Building

10.1 Model Selection

Logistic Regression with OneVsRestClassifier was chosen because:

- The task is **multi-label classification** — a customer can own multiple products
- OneVsRest trains one binary classifier per product (24 classifiers total)
- Logistic Regression outputs **probability scores** used as recommendation confidence
- Computationally efficient — well suited for federated simulation

10.2 Define the Flower Federated Client

Each `FinancialClient` represents one bank branch / age group.
It implements the Flower `NumPyClient` interface with 4 key methods:

Method	Purpose
`get_parameters()`	Extract model weights to send to server
`set_parameters()`	Receive aggregated global weights from server
`fit()`	Train locally on private data and return updated weights
`evaluate()`	Measure local performance with F1, Precision, Recall

```
class FinancialClient(fl.client.NumPyClient):

    def __init__(self, X_train, X_test, y_train, y_test, client_id):
        self.X_train = X_train
        self.X_test  = X_test
```

```
        self.y_train = y_train
        self.y_test  = y_test
        self.client_id = client_id

        # Each client has its own local model
        self.model = OneVsRestClassifier(
            LogisticRegression(
                                max_iter=500,
                                random_state=42)
        )
        # Fit once so model has coef_ attribute
        # This initial fit might result in _ConstantPredictor for some classes
        self.model.fit(self.X_train, self.y_train)

    def get_parameters(self, config):
        """Send model weights to server"""
        params = []
        n_features = self.X_train.shape[1] # Get number of features from X_train

        for estimator in self.model.estimators_:
            if hasattr(estimator, 'coef_') and hasattr(estimator, 'intercept_'):
                params.append(estimator.coef_.flatten())
                params.append(estimator.intercept_.flatten())
            else:
                # If it's a _ConstantPredictor, it doesn't have coef_ or intercept_.
                # Append zero arrays of the expected shape to maintain consistency.
                # coef_ shape: (1, n_features) -> flattened to (n_features,)
                # intercept_ shape: (1,) -> flattened to (1,)
                params.append(np.zeros(n_features, dtype=float).flatten())
                params.append(np.zeros(1, dtype=float).flatten())
        return params

    def set_parameters(self, parameters):
        """Receive aggregated weights from server"""
        idx = 0
        n_features = self.X_train.shape[1] # Get number of features from X_train

        for estimator in self.model.estimators_:
            if hasattr(estimator, 'coef_') and hasattr(estimator, 'intercept_'):
                # Reshape parameters back to original shape for coef_ and intercept_
                # coef_ should be (1, n_features) and intercept_ (1,)
                estimator.coef_ = parameters[idx].reshape((1, n_features))
                estimator.intercept_ = parameters[idx + 1].reshape((1,))
                idx += 2
            else:
                # If it's a _ConstantPredictor, we skip assigning parameters to it.
                # However, we must advance idx by the number of parameters it *would* have had
                # (i.e., 2, for coef_ and intercept_ placeholders) to keep parameter list indexing correct.
                idx += 2
                # A _ConstantPredictor will remain constant based on its original state.

    def fit(self, parameters, config):
        """Train locally and return updated weights"""
        self.set_parameters(parameters)
        self.model.fit(self.X_train, self.y_train)
        print(f"  Client {self.client_id} trained on {len(self.X_train)} samples")
        return self.get_parameters(config), len(self.X_train), {}

    def evaluate(self, parameters, config):
        """Evaluate local model and return metrics"""
        self.set_parameters(parameters)
        y_pred = self.model.predict(self.X_test)

        f1        = f1_score(self.y_test, y_pred, average='weighted', zero_division=0)
        precision = precision_score(self.y_test, y_pred, average='weighted', zero_division=0)
        recall    = recall_score(self.y_test, y_pred, average='weighted', zero_division=0)

        print(f"  Client {self.client_id} → F1: {f1:.4f} | "
              f"Precision: {precision:.4f} | Recall: {recall:.4f}")

        return float(f1), len(self.X_test), {
            "f1": float(f1),
            "precision": float(precision),
            "recall": float(recall)
        }

print("Flower client defined!")
```

```
Flower client defined!
```

10.3 Instantiate the 3 Client Objects

```
# Instantiate all 3 clients (re-running this to pick up the updated FinancialClient class definition)
client1 = FinancialClient(X1_train, X1_test, y1_train, y1_test, client_id=1)
client2 = FinancialClient(X2_train, X2_test, y2_train, y2_test, client_id=2)
client3 = FinancialClient(X3_train, X3_test, y3_train, y3_test, client_id=3)

print("All 3 clients created!")
```

```
All 3 clients created!
```

10.4 FedAvg Aggregation Algorithm

Federated Averaging (FedAvg) works as follows each round:

1. Each client trains locally on its **private** data
2. Clients send model **weights** (not data) to central server
3. Server computes a **weighted average** proportional to dataset size
4. Averaged global weights are sent back to all clients
5. Repeat for N rounds

```
def federated_average(clients, rounds=5):
    """
    Simulate FedAvg — the core federated learning algorithm.
    Each round:
      1. Each client trains locally
      2. Server collects weights
      3. Server averages weights (FedAvg)
      4. Server sends averaged weights back
    """
    print("=" * 50)
    print("  FEDERATED LEARNING — FedAvg Simulation")
    print("=" * 50)

    # Get initial parameters from client 1
    global_params = clients[0].get_parameters(config={})

    for round_num in range(1, rounds + 1):
        print(f"\n🔄 Round {round_num}/{rounds}")

        all_params = []
        all_sizes  = []

        # Step 1: Each client trains locally
        for client in clients:
            params, size, _ = client.fit(global_params, config={})
            all_params.append(params)
            all_sizes.append(size)

        # Step 2: FedAvg — weighted average of all client weights
        total = sum(all_sizes)
        global_params = [
            sum(all_params[c][i] * (all_sizes[c] / total)
                for c in range(len(clients)))
            for i in range(len(all_params[0]))
        ]

        # Step 3: Evaluate each client with global weights
        print(f"  📊 Evaluation after Round {round_num}:")
        for client in clients:
            client.evaluate(global_params, config={})

    print("\n" + "=" * 50)
    print("Federated Learning Complete!")
    print("=" * 50)
    return global_params

# Run federated learning for 5 rounds
global_params = federated_average(
```

```
    clients=[client1, client2, client3],
    rounds=5
)
```

```
==================================================
  FEDERATED LEARNING — FedAvg Simulation
==================================================

🔄 Round 1/5
  Client 1 trained on 30396 samples
  Client 2 trained on 7147 samples
  Client 3 trained on 2181 samples
  📊 Evaluation after Round 1:
  Client 1 → F1: 0.9528 | Precision: 0.9507 | Recall: 0.9571
  Client 2 → F1: 0.3927 | Precision: 0.7960 | Recall: 0.4320
  Client 3 → F1: 0.3558 | Precision: 0.4265 | Recall: 0.3960

🔄 Round 2/5
  Client 1 trained on 30396 samples
  Client 2 trained on 7147 samples
  Client 3 trained on 2181 samples
  📊 Evaluation after Round 2:
  Client 1 → F1: 0.9528 | Precision: 0.9507 | Recall: 0.9571
  Client 2 → F1: 0.3927 | Precision: 0.7960 | Recall: 0.4320
  Client 3 → F1: 0.3558 | Precision: 0.4265 | Recall: 0.3960

🔄 Round 3/5
  Client 1 trained on 30396 samples
  Client 2 trained on 7147 samples
  Client 3 trained on 2181 samples
  📊 Evaluation after Round 3:
  Client 1 → F1: 0.9528 | Precision: 0.9507 | Recall: 0.9571
  Client 2 → F1: 0.3927 | Precision: 0.7960 | Recall: 0.4320
  Client 3 → F1: 0.3558 | Precision: 0.4265 | Recall: 0.3960

🔄 Round 4/5
  Client 1 trained on 30396 samples
  Client 2 trained on 7147 samples
  Client 3 trained on 2181 samples
  📊 Evaluation after Round 4:
  Client 1 → F1: 0.9528 | Precision: 0.9507 | Recall: 0.9571
  Client 2 → F1: 0.3927 | Precision: 0.7960 | Recall: 0.4320
  Client 3 → F1: 0.3558 | Precision: 0.4265 | Recall: 0.3960

🔄 Round 5/5
  Client 1 trained on 30396 samples
  Client 2 trained on 7147 samples
  Client 3 trained on 2181 samples
  📊 Evaluation after Round 5:
  Client 1 → F1: 0.9528 | Precision: 0.9507 | Recall: 0.9571
  Client 2 → F1: 0.3927 | Precision: 0.7960 | Recall: 0.4320
  Client 3 → F1: 0.3558 | Precision: 0.4265 | Recall: 0.3960

==================================================
Federated Learning Complete!
==================================================
```

---

11. 🔧 Hyperparameter Tuning

We systematically test different model configurations to find optimal settings.
This answers **Q5 and Q6** from our research questions.

11.1 Tune Regularisation Strength (C)

`C` controls the trade-off between fitting the training data and generalising.
Lower C = stronger regularisation | Higher C = weaker regularisation

```
from sklearn.linear_model import LogisticRegression
from sklearn.multiclass import OneVsRestClassifier
from sklearn.metrics import f1_score, precision_score, recall_score
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# Try different C values (regularization strength)
# Lower C = more regularization, Higher C = less regularization
C_values = [0.01, 0.1, 1, 10, 100]
```

```
print("=" * 55)
print("  FINE TUNING — Logistic Regression (Client 2 & 3)")
print("=" * 55)

for C in C_values:
    print(f"\n🔗 Testing C = {C}")

    model = OneVsRestClassifier(
        LogisticRegression(C=C, max_iter=1000,
                           solver='saga',
                           random_state=42)
    )
    # Test on Client 1
    model.fit(X1_train, y1_train)
    y_pred = model.predict(X1_test)

    f1        = f1_score(y1_test, y_pred, average='weighted', zero_division=0)
    precision = precision_score(y1_test, y_pred, average='weighted', zero_division=0)
    recall    = recall_score(y1_test, y_pred, average='weighted', zero_division=0)

    print(f" Client 1 - F1: {f1:.4f} | Precision: {precision:.4f} | Recall: {recall:.4f}")

    # Test on Client 2 (weakest after client 3)
    model.fit(X2_train, y2_train)
    y_pred = model.predict(X2_test)

    f1        = f1_score(y2_test, y_pred, average='weighted', zero_division=0)
    precision = precision_score(y2_test, y_pred, average='weighted', zero_division=0)
    recall    = recall_score(y2_test, y_pred, average='weighted', zero_division=0)

    print(f"  Client 2 → F1: {f1:.4f} | Precision: {precision:.4f} | Recall: {recall:.4f}")

    # Test on Client 3
    model.fit(X3_train, y3_train)
    y_pred = model.predict(X3_test)

    f1        = f1_score(y3_test, y_pred, average='weighted', zero_division=0)
    precision = precision_score(y3_test, y_pred, average='weighted', zero_division=0)
    recall    = recall_score(y3_test, y_pred, average='weighted', zero_division=0)

    print(f"  Client 3 → F1: {f1:.4f} | Precision: {precision:.4f} | Recall: {recall:.4f}")
```

```
=======================================================
  FINE TUNING — Logistic Regression (Client 2 & 3)
=======================================================

🔗 Testing C = 0.01
 Client 1 - F1: 0.9316 | Precision: 0.9540 | Recall: 0.9255
  Client 2 → F1: 0.6286 | Precision: 0.7916 | Recall: 0.5832
  Client 3 → F1: 0.5248 | Precision: 0.7553 | Recall: 0.4967

🔗 Testing C = 0.1
 Client 1 - F1: 0.9326 | Precision: 0.9552 | Recall: 0.9261
  Client 2 → F1: 0.6542 | Precision: 0.7869 | Recall: 0.6110
  Client 3 → F1: 0.6269 | Precision: 0.7345 | Recall: 0.5945

🔗 Testing C = 1
 Client 1 - F1: 0.9326 | Precision: 0.9548 | Recall: 0.9261
  Client 2 → F1: 0.6594 | Precision: 0.7838 | Recall: 0.6183
  Client 3 → F1: 0.6356 | Precision: 0.7341 | Recall: 0.6058

🔗 Testing C = 10
 Client 1 - F1: 0.9326 | Precision: 0.9548 | Recall: 0.9261
  Client 2 → F1: 0.6591 | Precision: 0.7830 | Recall: 0.6183
  Client 3 → F1: 0.6353 | Precision: 0.7332 | Recall: 0.6058

🔗 Testing C = 100
 Client 1 - F1: 0.9326 | Precision: 0.9548 | Recall: 0.9261
  Client 2 → F1: 0.6593 | Precision: 0.7828 | Recall: 0.6186
  Client 3 → F1: 0.6353 | Precision: 0.7332 | Recall: 0.6058
```

11.2 Tune Optimisation Solver

Different solvers use different algorithms to minimise the loss function.
We test three options to confirm which gives the best results.

```
solvers = ['lbfgs', 'saga', 'liblinear']

print("=" * 55)
print("  FINE TUNING — Solver Comparison")
print("=" * 55)

for solver in solvers:
    print(f"\n🔗 Testing solver = {solver}")

    model = OneVsRestClassifier(
        LogisticRegression(C=1, max_iter=1000,
                           solver=solver,
                           random_state=42)
    )
    #Client 1
    model.fit(X1_train, y1_train)
    y_pred = model.predict(X1_test)
    f1 = f1_score(y1_test, y_pred, average='weighted', zero_division=0)
    print(f" Client 1  → F1: {f1:.4f}")

    # Client 2
    model.fit(X2_train, y2_train)
    y_pred = model.predict(X2_test)
    f1 = f1_score(y2_test, y_pred, average='weighted', zero_division=0)
    print(f"  Client 2 → F1: {f1:.4f}")

    # Client 3
    model.fit(X3_train, y3_train)
    y_pred = model.predict(X3_test)
    f1 = f1_score(y3_test, y_pred, average='weighted', zero_division=0)
    print(f"  Client 3 → F1: {f1:.4f}")
```

```
=======================================================
  FINE TUNING — Solver Comparison
=======================================================

🔗 Testing solver = lbfgs
 Client 1  → F1: 0.9326
  Client 2 → F1: 0.6594
  Client 3 → F1: 0.6356

🔗 Testing solver = saga
 Client 1  → F1: 0.9326
  Client 2 → F1: 0.6594
  Client 3 → F1: 0.6356

🔗 Testing solver = liblinear
 Client 1  → F1: 0.9326
  Client 2 → F1: 0.6590
  Client 3 → F1: 0.6360
```

11.3 Tune Class Weights

`class_weight='balanced'` automatically adjusts weights to handle product imbalance (most customers only own 1–2 products out of 24).

```
print("=" * 55)
print("  FINE TUNING — Class Weight Balancing")
print("=" * 55)

for weight in [None, 'balanced']:
    print(f"\n🔗 Testing class_weight = {weight}")

    model = OneVsRestClassifier(
        LogisticRegression(C=1, max_iter=1000,
                           class_weight=weight,
                           solver='saga',
                           random_state=42)
    )
    #client 1
    model.fit(X1_train, y1_train)
    y_pred = model.predict(X1_test)
    f1        = f1_score(y1_test, y_pred, average='weighted', zero_division=0)
    recall    = recall_score(y1_test, y_pred, average='weighted', zero_division=0)
    print(f" Client 1  → F1: {f1:.4f} | Recall: {recall:.4f}")
```

```
    # Client 2
    model.fit(X2_train, y2_train)
    y_pred = model.predict(X2_test)
    f1        = f1_score(y2_test, y_pred, average='weighted', zero_division=0)
    recall    = recall_score(y2_test, y_pred, average='weighted', zero_division=0)
    print(f"  Client 2 → F1: {f1:.4f} | Recall: {recall:.4f}")

    # Client 3
    model.fit(X3_train, y3_train)
    y_pred = model.predict(X3_test)
    f1        = f1_score(y3_test, y_pred, average='weighted', zero_division=0)
    recall    = recall_score(y3_test, y_pred, average='weighted', zero_division=0)
    print(f"  Client 3 → F1: {f1:.4f} | Recall: {recall:.4f}")
```

```
======================================================
  FINE TUNING — Class Weight Balancing
======================================================

🔗 Testing class_weight = None
 Client 1  → F1: 0.9326 | Recall: 0.9261
  Client 2 → F1: 0.6594 | Recall: 0.6183
  Client 3 → F1: 0.6356 | Recall: 0.6058

🔗 Testing class_weight = balanced
 Client 1  → F1: 0.8028 | Recall: 0.7252
  Client 2 → F1: 0.6118 | Recall: 0.5979
  Client 3 → F1: 0.4255 | Recall: 0.4600
```

11.4 Tuning Summary

Parameter	Options Tested	Best Value	Reason
C (regularisation)	0.01, 0.1, 1, 10, 100	**1**	Sweet spot — no gain beyond this
solver	lbfgs, saga, liblinear	**lbfgs**	Fastest with same accuracy
class_weight	None, balanced	**None**	Balanced reduced F1 for all clients
max_iter	500 → 1000	**1000**	Ensures full model convergence

Improvement from tuning:

Client	Before Tuning	After Tuning	Gain
Client 1 (Young)	0.9528	0.9489	Stable (reduced overfitting)
Client 2 (Middle)	0.3927	0.7036	**+0.31** 🎉
Client 3 (Older)	0.3558	0.6795	**+0.32** 🎉

---

12. Final Model — Tuned Federated Learning

Run the final federated learning simulation using the best hyperparameters:

- `C=1`, `solver='lbfgs'`, `max_iter=1000`, `class_weight=None`

Constant product columns (all 0s or all 1s) are removed first to prevent `_ConstantPredictor` errors which would break the weight sharing mechanism.

```
# Find and remove product columns with no variation
product_cols_filtered = []
for c in product_cols:
    unique_vals = df_pandas[c].nunique()
    if unique_vals > 1:
        product_cols_filtered.append(c)
    else:
        print(f"Removed constant column: {c}")

print(f"\nOriginal products : {len(product_cols)}")
print(f"Filtered products : {len(product_cols_filtered)}")

# Re-extract X and y with filtered products
y1 = client_1[product_cols_filtered].values
y2 = client_2[product_cols_filtered].values
y3 = client_3[product_cols_filtered].values

# Re-prepare clients
X1_train, X1_test, y1_train, y1_test = prepare_client(X1, y1)
X2_train, X2_test, y2_train, y2_test = prepare_client(X2, y2)
X3_train, X3_test, y3_train, y3_test = prepare_client(X3, y3)
```

```
print("\n Data re-prepared with filtered products!")
```

```
Removed constant column: has_savings_account
Removed constant column: has_guarantees
Removed constant column: has_derivada_account
Removed constant column: has_particular_account
Removed constant column: has_particular_plus_account

Original products : 24
Filtered products : 19

 Data re-prepared with filtered products!
```

```
class FinancialClient(fl.client.NumPyClient):

    def __init__(self, X_train, X_test, y_train, y_test, client_id):
        self.X_train   = X_train
        self.X_test    = X_test
        self.y_train   = y_train
        self.y_test    = y_test
        self.client_id = client_id
        self.model     = OneVsRestClassifier(
            LogisticRegression(C=1, solver='lbfgs',
                               max_iter=1000, random_state=42)
        )
        self.model.fit(self.X_train, self.y_train)

    def get_parameters(self, config):
        params = []
        for est in self.model.estimators_:
            if hasattr(est, 'coef_'):
                params.append(est.coef_.flatten())
                params.append(est.intercept_.flatten())
            else:
                # Handle constant predictor safely
                params.append(np.zeros(self.X_train.shape[1]))
                params.append(np.zeros(1))
        return params

    def set_parameters(self, parameters):
        idx = 0
        for est in self.model.estimators_:
            if hasattr(est, 'coef_'):
                est.coef_      = parameters[idx].reshape(est.coef_.shape)
                idx += 1
                est.intercept_ = parameters[idx].reshape(est.intercept_.shape)
                idx += 1
            else:
                idx += 2

    def fit(self, parameters, config):
        self.model.fit(self.X_train, self.y_train)
        print(f"  Client {self.client_id} trained on {len(self.X_train)} samples")
        return self.get_parameters(config), len(self.X_train), {}

    def evaluate(self, parameters, config):
        y_pred    = self.model.predict(self.X_test)
        f1        = f1_score(self.y_test, y_pred, average='weighted', zero_division=0)
        precision = precision_score(self.y_test, y_pred, average='weighted', zero_division=0)
        recall    = recall_score(self.y_test, y_pred, average='weighted', zero_division=0)
        print(f"  Client {self.client_id} → F1: {f1:.4f} | "
              f"Precision: {precision:.4f} | Recall: {recall:.4f}")
        return float(f1), len(self.X_test), {
            "f1": float(f1),
            "precision": float(precision),
            "recall": float(recall)
        }

# Re-create and run
client1 = FinancialClient(X1_train, X1_test, y1_train, y1_test, client_id=1)
client2 = FinancialClient(X2_train, X2_test, y2_train, y2_test, client_id=2)
client3 = FinancialClient(X3_train, X3_test, y3_train, y3_test, client_id=3)

global_params = federated_average(
    clients=[client1, client2, client3],
```

```
    rounds=5
)
```

```
==================================================
  FEDERATED LEARNING — FedAvg Simulation
==================================================

🔄 Round 1/5
  Client 1 trained on 30396 samples
  Client 2 trained on 7147 samples
  Client 3 trained on 2181 samples
  📊 Evaluation after Round 1:
  Client 1 → F1: 0.9489 | Precision: 0.9542 | Recall: 0.9493
  Client 2 → F1: 0.7036 | Precision: 0.7495 | Recall: 0.7018
  Client 3 → F1: 0.6795 | Precision: 0.7165 | Recall: 0.6839

🔄 Round 2/5
  Client 1 trained on 30396 samples
  Client 2 trained on 7147 samples
  Client 3 trained on 2181 samples
  📊 Evaluation after Round 2:
  Client 1 → F1: 0.9489 | Precision: 0.9542 | Recall: 0.9493
  Client 2 → F1: 0.7036 | Precision: 0.7495 | Recall: 0.7018
  Client 3 → F1: 0.6795 | Precision: 0.7165 | Recall: 0.6839

🔄 Round 3/5
  Client 1 trained on 30396 samples
  Client 2 trained on 7147 samples
  Client 3 trained on 2181 samples
  📊 Evaluation after Round 3:
  Client 1 → F1: 0.9489 | Precision: 0.9542 | Recall: 0.9493
  Client 2 → F1: 0.7036 | Precision: 0.7495 | Recall: 0.7018
  Client 3 → F1: 0.6795 | Precision: 0.7165 | Recall: 0.6839

🔄 Round 4/5
  Client 1 trained on 30396 samples
  Client 2 trained on 7147 samples
  Client 3 trained on 2181 samples
  📊 Evaluation after Round 4:
  Client 1 → F1: 0.9489 | Precision: 0.9542 | Recall: 0.9493
  Client 2 → F1: 0.7036 | Precision: 0.7495 | Recall: 0.7018
  Client 3 → F1: 0.6795 | Precision: 0.7165 | Recall: 0.6839

🔄 Round 5/5
  Client 1 trained on 30396 samples
  Client 2 trained on 7147 samples
  Client 3 trained on 2181 samples
  📊 Evaluation after Round 5:
  Client 1 → F1: 0.9489 | Precision: 0.9542 | Recall: 0.9493
  Client 2 → F1: 0.7036 | Precision: 0.7495 | Recall: 0.7018
  Client 3 → F1: 0.6795 | Precision: 0.7165 | Recall: 0.6839

==================================================
Federated Learning Complete!
==================================================
```

13. Model Evaluation

Metric Definitions

- **Precision** — Of all products recommended, what fraction were correct?
- **Recall** — Of all products a customer actually owns, what fraction did we predict?
- **F1 Score** — Harmonic mean of Precision and Recall (our primary metric)

Final Results (Q5 & Q6)

Client	Group	Samples	F1 Score	Precision	Recall
Client 1	Young (≤30)	30,396	**0.9489**	0.9542	0.9493
Client 2	Middle (30-50)	7,147	**0.7036**	0.7495	0.7018
Client 3	Older (>50)	2,181	**0.6795**	0.7165	0.6839

**All clients exceed the F1 > 0.70 success criterion **

Answer to Q6: Client 1 performs best because it has the most training data (~30k samples).
Clients 2 and 3 improved dramatically (+0.31, +0.32) after hyperparameter tuning.

Answer to Q5: Yes — federated learning successfully trained a model that achieves strong recommendation accuracy while keeping raw data private across all 3 clients.

---

14. Model Saving

We save **3 files** needed for deployment using **Joblib** — the recommended format for Scikit-learn models:

File	Purpose
`best_model_client1.joblib`	Trained model that makes predictions
`scaler.joblib`	Feature scaler — must match training scaling
`product_cols.joblib`	Product names — needed to label recommendations

Client 1 is chosen as the deployment model because it has the highest F1 (0.9489) and was trained on the largest dataset (30,396 customers).

```
import joblib
import os
from sklearn.preprocessing import StandardScaler # Import StandardScaler

# Create save directory (corrected to 'federated_models' for consistency)
os.makedirs('federated_models', exist_ok=True)

# Save Client 1 model (best performer — F1: 0.9489)
joblib.dump(client1.model, 'federated_models/best_model_client1.joblib')

# Fix: Create and fit a scaler to be saved.
# We use X1 (unscaled features for client 1) to fit the scaler,
# as it represents the input data distribution.
scaler = StandardScaler()
scaler.fit(X1) # Fit on the unscaled features of client 1

# Save scaler (MUST save this too for predictions to work)
joblib.dump(scaler, 'federated_models/scaler.joblib')

# Save product column names (needed for recommendations)
joblib.dump(product_cols_filtered, 'federated_models/product_cols.joblib')

# Verify files saved correctly
for f in os.listdir('federated_models'):
    size = os.path.getsize(f'federated_models/{f}')
    print(f" {f} — {size/1024:.1f} KB")
```

```
✅ product_cols.joblib — 0.4 KB
✅ best_model_client1.joblib — 8.1 KB
✅ scaler.joblib — 0.8 KB
```

---

15. Deployment — Gradio User Interface

A customer-facing financial advisor UI built with **Gradio**.

The interface allows any bank customer to:

1. Enter personal details (age, income, seniority, employment)
2. Select their reason for visiting the bank
3. Receive top 5 personalised financial product recommendations
4. View a confidence score for each recommendation
5. Get clear next steps on how to sign up

This completes the full pipeline from raw data → trained model → live recommendations.

How the Recommendation Works

1. Customer inputs their profile
2. Profile is scaled using the saved `StandardScaler`
3. The model predicts probability for each of the 24 products
4. Products are filtered by the customer's stated goal
5. Top 5 products are displayed with confidence scores and explanations

```
import gradio as gr
import joblib
import numpy as np

# ── Load saved model files ──
model                 = joblib.load('federated_models/best_model_client1.joblib')
scaler                = joblib.load('federated_models/scaler.joblib')
product_cols_filtered = joblib.load('federated_models/product_cols.joblib')

# ── Friendly product descriptions ──
product_info = {
    'has_savings_account'        : {
        'name'    : '💰 Savings Account',
        'desc'    : 'Grow your money safely with competitive interest rates.',
        'best_for': 'Anyone looking to save for the future'
    },
    'has_current_account'        : {
        'name'    : '🏦 Current Account',
        'desc'    : 'Your everyday account for payments, transfers and direct debits.',
        'best_for': 'Everyone — the foundation of your banking'
    },
    'has_credit_card'            : {
        'name'    : '💳 Credit Card',
        'desc'    : 'Buy now, pay later with flexible repayment options.',
        'best_for': 'Professionals who want spending flexibility'
    },
    'has_mortgage'               : {
        'name'    : '🏠 Mortgage',
        'desc'    : 'Finance your dream home with competitive mortgage rates.',
        'best_for': 'Those ready to buy their first or next home'
    },
    'has_pension_plan'           : {
        'name'    : '🤑 Pension Plan',
        'desc'    : 'Start building your retirement fund today.',
        'best_for': 'Anyone thinking about long-term financial security'
    },
    'has_pension'                : {
        'name'    : '🤑 Pension',
        'desc'    : 'Secure monthly income when you retire.',
        'best_for': 'Customers planning for retirement'
    },
    'has_loan'                   : {
        'name'    : '🆘 Personal Loan',
        'desc'    : 'Get funds quickly for any personal need or project.',
        'best_for': 'Those needing a financial boost'
    },
    'has_funds'                  : {
        'name'    : '📈 Investment Funds',
        'desc'    : 'Grow your wealth by investing in diversified funds.',
        'best_for': 'Customers ready to start investing'
    },
    'has_securities'             : {
        'name'    : '📉 Stocks and Securities',
        'desc'    : 'Trade and invest in stocks and financial securities.',
        'best_for': 'Experienced investors seeking higher returns'
    },
    'has_eaccount'               : {
        'name'    : '💻 Online Account',
        'desc'    : 'A fully digital account — manage everything from your phone.',
        'best_for': 'Tech-savvy customers who prefer online banking'
    },
    'has_payroll_account'        : {
        'name'    : '💼 Payroll Account',
        'desc'    : 'Receive your salary directly and enjoy exclusive benefits.',
        'best_for': 'Employed customers wanting salary management'
    },
    'has_payroll'                : {
        'name'    : '🆘 Payroll Service',
        'desc'    : 'Link your salary to unlock premium banking benefits.',
        'best_for': 'Salaried employees'
    },
    'has_direct_debit'           : {
        'name'    : '🔄 Direct Debit',
        'desc'    : 'Automate your bill payments — never miss a payment again.',
        'best_for': 'Anyone who wants hassle-free bill management'
    },
    'has_short_term_deposit'     : {
```

```
        'name'    : '📅 Short-Term Deposit',
        'desc'    : 'Lock in your money for a short period and earn higher interest.',
        'best_for': 'Savers with money to spare for 1-6 months'
    },
    'has_medium_term_deposit'    : {
        'name'    : '🗓️ Medium-Term Deposit',
        'desc'    : 'Earn better returns by saving for 6-24 months.',
        'best_for': 'Customers with medium-term savings goals'
    },
    'has_long_term_deposit'      : {
        'name'    : '1️⃣ Long-Term Deposit',
        'desc'    : 'Maximise your returns by committing to a long-term savings plan.',
        'best_for': 'Customers with long-term financial goals'
    },
    'has_junior_account'         : {
        'name'    : '🎓 Junior Account',
        'desc'    : 'A specially designed account for students and young people.',
        'best_for': 'Students and young adults under 25'
    },
    'has_home_account'           : {
        'name'    : '🏠 Home Account',
        'desc'    : 'A dedicated account to help you save towards buying a home.',
        'best_for': 'Future homeowners saving for a deposit'
    },
    'has_taxes'                  : {
        'name'    : '📄 Tax Management Account',
        'desc'    : 'Simplify your tax payments and obligations.',
        'best_for': 'Self-employed or business customers'
    },
    'has_guarantees'             : {
        'name'    : '🔒 Bank Guarantee',
        'desc'    : 'Use the bank as a guarantor for contracts or rentals.',
        'best_for': 'Business owners or those renting property'
    },
    'has_derivada_account'       : {
        'name'    : '📊 Derivada Account',
        'desc'    : 'A flexible account linked to market performance.',
        'best_for': 'Customers interested in market-linked savings'
    },
    'has_particular_account'     : {
        'name'    : '👤 Personal Account',
        'desc'    : 'A tailored account designed for individual needs.',
        'best_for': 'Individual customers seeking personalised banking'
    },
    'has_particular_plus_account': {
        'name'    : '👤 Personal Plus Account',
        'desc'    : 'An upgraded personal account with additional benefits.',
        'best_for': 'Customers wanting premium personal banking'
    },
    'has_mas_particular_account' : {
        'name'    : '⭐ Premium Personal Account',
        'desc'    : 'Our top-tier personal account with exclusive perks.',
        'best_for': 'High-value customers seeking the best service'
    },
}

# ── Helper functions ──
def get_life_stage(age):
    if age <= 22:
        return "Student", "🎓"
    elif age <= 30:
        return "Young Professional", "💼"
    elif age <= 45:
        return "Established Adult", "🏠"
    elif age <= 60:
        return "Pre-Retirement", "📊"
    else:
        return "Senior", "🌅"

def get_income_bracket(income):
    if income < 20000:
        return "Entry Level"
    elif income < 50000:
        return "Middle Income"
    elif income < 100000:
        return "Upper Middle"
    else:
```

```
        return "High Income"

def get_confidence_summary(top5):
    top5_probs   = [prob for _, prob in top5]
    overall_conf = np.mean(top5_probs) * 100
    highest_conf = max(top5_probs) * 100
    lowest_conf  = min(top5_probs) * 100
    if overall_conf >= 60:
        rating = "🟢 HIGH — Very confident in these recommendations"
    elif overall_conf >= 30:
        rating = "🟡 MEDIUM — Good recommendations for your profile"
    else:
        rating = "🔵 FAIR — Based on limited profile information"
    return overall_conf, highest_conf, lowest_conf, rating

# ── Main recommendation function ──
def advise_customer(age, income, seniority, employment, reason_for_visit):

    # Map inputs to model features
    is_new         = 1 if seniority == 0 else 0
    is_active      = 1
    gender_idx     = 0
    segment_idx    = 2 if age <= 25 else (0 if income > 100000 else 1)
    is_high_earner = 1 if income > 150000 else 0
    total_products = 0

    profile        = [age, income, seniority, is_active, is_new,
                      gender_idx, segment_idx, total_products, is_high_earner]
    profile_scaled = scaler.transform([profile])

    # Get probabilities for all products
    probabilities = []
    for i, est in enumerate(model.estimators_):
        prob = est.predict_proba(profile_scaled)[0][1] \
               if hasattr(est, 'predict_proba') else 0.0
        probabilities.append((product_cols_filtered[i], prob))

    probabilities.sort(key=lambda x: x[1], reverse=True)

    # Filter by reason for visit
    reason_filters = {
        'I want to save money'         : ['savings', 'deposit', 'funds'],
        'I want to buy a home'         : ['mortgage', 'home'],
        'I need a loan or credit'      : ['loan', 'credit_card'],
        'I want to invest'             : ['funds', 'securities'],
        'I am starting a new job'      : ['payroll', 'current', 'direct_debit'],
        'I am a student'               : ['junior', 'eaccount', 'current'],
        'I want to plan for retirement': ['pension', 'funds', 'securities'],
        'I just want general advice'   : []
    }

    keywords = reason_filters.get(reason_for_visit, [])
    if keywords:
        priority    = [(p, prob) for p, prob in probabilities
                       if any(k in p for k in keywords)]
        others      = [(p, prob) for p, prob in probabilities
                       if not any(k in p for k in keywords)]
        sorted_recs = priority + others
    else:
        sorted_recs = probabilities

    top5 = sorted_recs[:5]

    # Life stage and income
    stage_name, stage_emoji = get_life_stage(age)
    income_bracket          = get_income_bracket(income)

    # Confidence scores
    overall_conf, highest_conf, lowest_conf, rating = \
        get_confidence_summary(top5)

    # ────────────────────────────────────────
    # Panel 1 — Welcome & Profile
    # ────────────────────────────────────────
    welcome  = " WELCOME TO SANTANDER BANK\n"
    welcome += "─" * 45 + "\n\n"
    welcome += "  Hello! Based on your profile:\n\n"
    welcome += f"  {stage_emoji} Life Stage   : {stage_name}\n"
```

```
welcome += f"   Income      : EUR {income:,.0f} ({income_bracket})\n"
welcome += f"   With Us     : "
welcome += "New Customer\n" if seniority == 0 else f"{seniority} months\n"
welcome += f"   Employment  : {employment}\n\n"
welcome += f"   Your Goal   : {reason_for_visit}\n\n"
welcome += "−" * 45 + "\n"
welcome += "  Our AI has analysed thousands of\n"
welcome += "  customers just like you to find\n"
welcome += "  the best products for your needs.\n\n"
welcome += f"  Overall Match Confidence : {overall_conf:.1f}%\n"
welcome += f"  Rating : {rating}\n"

# ────────────────────────────────────
# Panel 2 — Top Recommendation
# ────────────────────────────────────
top_product = top5[0][0]
top_conf    = top5[0][1] * 100
top_info    = product_info.get(top_product, {
    'name': top_product, 'desc': '', 'best_for': ''
})
conf_bar    = "" * int(top_conf / 5) + "" * (20 - int(top_conf / 5))

top_rec  = " OUR TOP RECOMMENDATION FOR YOU\n"
top_rec += "−" * 45 + "\n\n"
top_rec += f"  {top_info['name']}\n\n"
top_rec += f"   What it is:\n"
top_rec += f"     {top_info['desc']}\n\n"
top_rec += f"  Best for:\n"
top_rec += f"     {top_info['best_for']}\n\n"
top_rec += "   Why we recommend this for you:\n"
if age <= 25:
    top_rec += "     As a young customer just starting out,\n"
    top_rec += "     this is the perfect first step.\n\n"
elif age <= 40:
    top_rec += "     At your stage of life, this product\n"
    top_rec += "     matches your financial goals perfectly.\n\n"
elif age <= 55:
    top_rec += "     With your experience and income level,\n"
    top_rec += "     this product will help you grow further.\n\n"
else:
    top_rec += "     As you plan for the future, this product\n"
    top_rec += "     offers the security you need.\n\n"
top_rec += f"   Confidence Score : {top_conf:.1f}%\n"
top_rec += f"     [{conf_bar}]\n"

# ────────────────────────────────────
# Panel 3 — Full Product Plan
# ────────────────────────────────────
all_recs  = "💡 YOUR PERSONALISED PRODUCT PLAN\n"
all_recs += "−" * 45 + "\n\n"
all_recs += "  Products we recommend based on\n"
all_recs += "  your profile and goals:\n\n"

for rank, (product, prob) in enumerate(top5, 1):
    info     = product_info.get(product, {
        'name': product, 'desc': 'Financial product', 'best_for': ''
    })
    pct      = prob * 100
    bar      = "" * int(pct / 5) + "" * (20 - int(pct / 5))
    priority = " Best Match"  if rank == 1 else \
               " Recommended" if rank <= 3 else \
               " Also Consider"
    all_recs += f"  {rank}. {info['name']}\n"
    all_recs += f"     {info['desc']}\n"
    all_recs += f"     Confidence : {pct:.1f}%  [{bar}]\n"
    all_recs += f"     {priority}\n\n"

# ────────────────────────────────────
# Panel 4 — Confidence Report
# ────────────────────────────────────
conf_out  = "📊 RECOMMENDATION CONFIDENCE REPORT\n"
conf_out += "−" * 45 + "\n\n"
conf_out += f"  Overall Confidence : {overall_conf:.1f}%\n"
conf_out += f"  Highest Match      : {highest_conf:.1f}%\n"
conf_out += f"  Lowest Match       : {lowest_conf:.1f}%\n\n"
conf_out += f"  Rating : {rating}\n\n"
conf_out += "−" * 45 + "\n"
```

```
    conf_out += "  SCORE BREAKDOWN\n\n"

    for product, prob in top5:
        info  = product_info.get(product, {'name': product})
        pct   = prob * 100
        bar   = "▓" * int(pct / 5) + "▒" * (20 - int(pct / 5))
        level = " Strong" if pct > 60 else \
                " Good"   if pct > 30 else \
                " Fair"   if pct > 10 else \
                " Low"
        conf_out += f"  {info['name']}\n"
        conf_out += f"  [{bar}] {pct:.1f}%  {level}\n\n"

    conf_out += "─" * 45 + "\n"
    conf_out += "  Confidence reflects how strongly\n"
    conf_out += "  the AI matched your profile to\n"
    conf_out += "  customers who benefited from\n"
    conf_out += "  each product.\n"

    # ────────────────────────────────────
    # Panel 5 — Next Steps
    # ────────────────────────────────────
    next_steps  = "📋 WHAT TO DO NEXT\n"
    next_steps += "─" * 45 + "\n\n"
    next_steps += "  Ready to get started? Here is how:\n\n"
    next_steps += "  1  Visit your nearest Santander branch\n"
    next_steps += "     with valid ID and proof of income\n\n"
    next_steps += "  2  Or apply online at santander.com\n"
    next_steps += "     in under 10 minutes\n\n"
    next_steps += "  3  Speak to one of our advisors\n"
    next_steps += "     for a free consultation\n\n"
    next_steps += "  Call us  : 0800-SANTANDER\n"
    next_steps += "  Web      : www.santander.com\n\n"
    next_steps += "─" * 45 + "\n"
    next_steps += "  Your data is 100% private.\n"
    next_steps += "  This recommendation was generated\n"
    next_steps += "  using Federated Learning — your\n"
    next_steps += "  personal data never left your device.\n"

    return welcome, top_rec, all_recs, conf_out, next_steps


# ── Build Gradio UI ──
with gr.Blocks(title="Santander Financial Advisor",
               theme=gr.themes.Soft()) as app:

    gr.Markdown("# Santander Smart Financial Advisor")
    gr.Markdown(
        "### Tell us about yourself and we will recommend "
        "the best products for your needs"
    )
    gr.Markdown("---")

    with gr.Row():
        with gr.Column():
            gr.Markdown("### 👤 About You")
            age       = gr.Slider(18, 80, value=25, step=1,
                                  label="How old are you?")
            income    = gr.Slider(0, 300000, value=35000, step=1000,
                                  label="What is your annual income? (EUR)")
            seniority = gr.Slider(0, 300, value=0, step=1,
                                  label="How long have you been with us? "
                                        "(0 = New Customer, in months)")
        with gr.Column():
            gr.Markdown("### 🎯 Your Situation")
            employment = gr.Dropdown(
                choices=['Employed Full-Time', 'Employed Part-Time',
                         'Self-Employed', 'Student',
                         'Retired', 'Unemployed'],
                value='Employed Full-Time',
                label="What is your employment status?"
            )
            reason = gr.Dropdown(
                choices=[
                    'I just want general advice',
                    'I want to save money',
                    'I want to buy a home',
                    'I need a loan or credit'
```

```
 'I need a loan or credit',
 'I want to invest',
 'I am starting a new job',
 'I am a student',
```